

A PROJECT REPORT
on
**HYBRID AEROFOIL DESIGN FOR ENHANCED STABILITY
IN TAILLESS UAVS**

submitted by

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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the Degree

of

Bachelor of Technology

in

Aeronautical Engineering



DEPARTMENT OF AERONAUTICAL ENGINEERING

ACE COLLEGE OF ENGINEERING

Thiruvallam, Thiruvananthapuram -695027

2025-2026

DECLARATION

I undersigned hereby declare that the project report “HYBRID AEROFOIL DESIGN FOR ENHANCED STABILITY IN TAILLESS UAVS”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Mr. Anand C L, Assistant Professor, Department of Aeronautical Engineering. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data idea fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed as the basis for the award of any degree, diploma, or similar title of any other University.

TRIVANDRUM

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CERTIFICATE

This is to certify that the report entitled "**HYBRID AEROFOIL DESIGN FOR ENHANCED STABILITY IN TAILLESS UAVS**" submitted by **ANUSREE M A, LAKSHMI ASWATHIHA S A, NITHIN G KRISHNAN** and **PRANAV ROY** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in AERONAUTICAL ENGINEERING during the academic year 2025-2026, is a Bonafide record of the project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

In the accomplishment of our project, we would like to express our deepest and sincere gratitude to **Mr. Anand C L**, Assistant Professor, Department of Aeronautical Engineering who guided us throughout the project. His sincerity, perseverance, and at most care have been a constant source of inspiration to us. We are extremely fortunate to have him as our internal guide. We extend our humble thanks to **Dr. Farrukh Sayeed**, Principal ACE College of Engineering, Trivandrum, Assistant Professor, **Sabu S. Joseph**, Head of the Department of Aeronautical Engineering, Assistant Professor, **Praveen J Sathyan**, Project Coordinator, Department of Aeronautical Engineering for their kind cooperation. We would also like to express our gratitude with a word of thanks to all of those who were directly or indirectly associated with the present research. It gives us pleasure to express our deepest thanks to our college library and librarian for their valuable assistance and help. We wish to thank our parents for their love, support, and encouragement throughout our study. We are also thankful to our friends whose enthusiasm gave wings to many ideas and we express our sincere thanks to everyone who helped us to make this project. Last but not least we thank the Almighty God for giving us the required vision and knowledge to complete this project work.

ABSTRACT

Airfoil geometry plays a crucial role in determining the aerodynamic performance and stability of unmanned aerial vehicles (UAVs). Conventional cambered airfoils typically generate high lift but produce a significant nose-down pitching moment, which requires additional control effort for stability. In contrast, reflex airfoils are designed to reduce the pitching moment and are therefore commonly used in tailless and flying-wing configurations. However, reflex airfoils generally produce lower lift compared to conventional airfoils, creating a trade-off between lift and stability.

This project investigates the aerodynamic characteristics of hybrid airfoils formed by blending the cambered NACA 2412 airfoil with the reflex MH60 airfoil. A total of nine hybrid configurations were generated by varying the geometric blending ratio from 90:10 to 10:90. Computational Fluid Dynamics (CFD) simulations were performed using ANSYS Fluent to evaluate aerodynamic performance.

Initially, all configurations were analyzed at an angle of attack of 4° , representing a typical cruise condition for UAVs. Based on these results, three representative configurations were selected for further analysis. These selected airfoils were then evaluated over a range of angles of attack from -2° to 8° to study their aerodynamic behaviour and longitudinal stability characteristics.

Key aerodynamic parameters including lift coefficient (C_l), drag coefficient (C_d), and pitching moment coefficient (C_m) were obtained for each case. The variation of C_m with angle of attack was analyzed to assess stability, while lift-to-drag ratio (C_l/C_d) was used to evaluate aerodynamic efficiency. The results show that increasing the proportion of reflex airfoil significantly reduces the pitching moment, improving stability, while moderate blending provides a better balance between lift and efficiency.

The study demonstrates that hybrid airfoils can effectively combine the high-lift characteristics of cambered airfoils with the improved stability of reflex airfoils. An optimal configuration was identified that provides a balanced trade-off between aerodynamic efficiency and longitudinal stability, making it suitable for UAV applications.

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CHAPTER 1

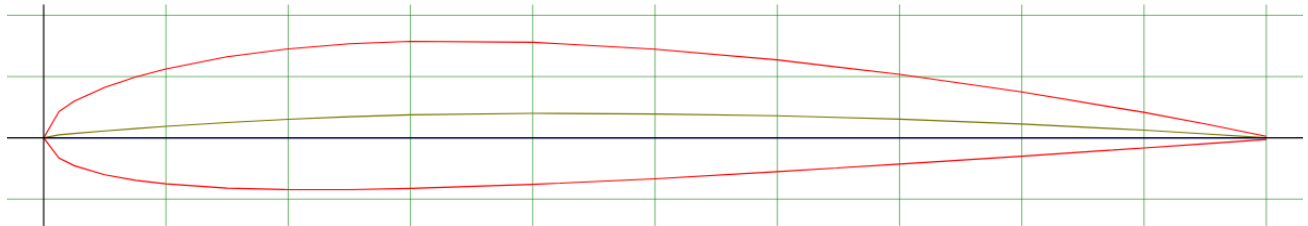
INTRODUCTION

The rapid advancement of Unmanned Aerial Vehicles (UAVs) has driven the need for efficient, stable, and aerodynamically optimized designs. The aerodynamic performance of a UAV largely depends on its aerofoil selection, which determines lift, drag, and stability characteristics. Conventional cambered aerofoils generate high lift but often exhibit negative pitching moments, leading to longitudinal instability. On the other hand, reflex aerofoils, characterized by an upward curvature near the trailing edge, produce a stabilizing positive pitching moment. This makes them ideal for flying-wing and tailless UAV configurations where tail surfaces are absent or minimized. However, reflex aerofoils typically generate less lift than their conventional counterparts, posing a challenge in applications that demand high aerodynamic efficiency. This research addresses this trade-off by designing a hybrid aerofoil geometry that blends the characteristics of both reflex and cambered aerofoils. The idea is to achieve a balance between lift and stability through geometric blending techniques and subsequent CFD-based aerodynamic analysis. The resulting configuration could offer UAVs better performance in terms of energy efficiency and flight stability, particularly in autonomous missions.

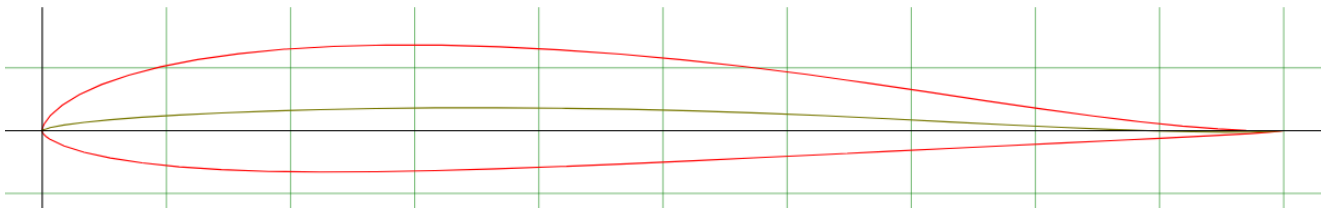
1.1 IMPORTANCE OF STABILITY IN UAVS

Stability plays a critical role in the design and performance of Unmanned Aerial Vehicles (UAVs), as it directly influences controllability, flight safety, and overall aerodynamic efficiency. A stable UAV can maintain its attitude and trajectory with minimal pilot or control input, which is essential for autonomous missions such as surveillance, mapping, and environmental monitoring. Longitudinal stability, in particular, ensures that the UAV can recover from small disturbances in pitch without excessive oscillations or control effort. Traditional cambered airfoils, while offering high lift, often produce a strong nose-down pitching moment that compromises this stability. In contrast, reflex airfoils generate a balancing nose-up moment, improving the static margin but at the cost of reduced lift. The hybrid airfoil concept effectively merges the strengths of both combining the lift-producing characteristics of a cambered section with the stability-inducing properties of a

reflex profile. This balanced aerodynamic behavior enables the UAV to achieve efficient flight performance with improved longitudinal stability, reduced trim drag, and enhanced maneuverability. Consequently, hybrid airfoils present a promising design alternative for UAVs requiring both endurance and stability under varying flight conditions.



1.1.1 NACA 2412 Airfoil



1.1.2 MH 60 Airfoil

1.2 OBJECTIVE OF THE STUDY

The primary objective of this study is to investigate the aerodynamic characteristics of hybrid airfoils formed by blending a conventional cambered airfoil with a reflex airfoil. The goal is to determine whether such hybrid airfoil configurations can provide a balanced combination of aerodynamic performance and longitudinal stability for UAV applications.

The specific objectives of the study are:

- To generate hybrid airfoil geometries by blending the cambered NACA 2412 airfoil with the reflex MH 60 airfoil in different geometric ratios.
- To perform Computational Fluid Dynamics (CFD) simulations using ANSYS Fluent to analyze the aerodynamic behavior of the generated hybrid airfoils.
- To evaluate key aerodynamic parameters such as lift coefficient (C_l), drag coefficient (C_d), and pitching moment coefficient (C_m).
- To analyze the effect of blending ratio on aerodynamic performance and pitching moment characteristics.
- To identify an optimal hybrid airfoil configuration suitable for UAV stability and performance.

1.3 METHODOLOGICAL APPROACH

The present study adopts a computational approach to evaluate the aerodynamic characteristics of hybrid airfoils. Initially, two base airfoils were selected: the cambered NACA 2412 airfoil and the reflex MH 60 airfoil. These airfoils were chosen due to their widely known aerodynamic characteristics and their contrasting pitching moment behaviors.

Hybrid airfoils were generated by blending the geometries of the two airfoils in different ratios. A total of nine hybrid airfoil configurations were created with blending ratios ranging from 90:10 to 10:90, representing varying contributions from the cambered and reflex airfoils.

The aerodynamic performance of these hybrid airfoils was analyzed using Computational Fluid Dynamics (CFD) simulations in ANSYS Fluent. A two-dimensional computational domain was created around each airfoil profile, and an appropriate mesh was generated to capture the flow behavior accurately. The simulations were conducted at a fixed angle of attack of 4° , which

represents a typical operating condition for small UAVs. The simulations produced aerodynamic coefficients such as lift coefficient (C_l), drag coefficient (C_d), and pitching moment coefficient (C_m), which were then compared for different hybrid airfoil configurations to understand the effect of blending ratio on aerodynamic performance and stability.

1.4 RESEARCH GAP

Despite extensive studies on aerofoil aerodynamics and UAV stability, there remains a noticeable gap in understanding how hybrid reflex camber aerofoils can be optimized to balance both aerodynamic efficiency and static longitudinal stability. Conventional cambered aerofoils generally enhance lift but tend to compromise pitch stability, while reflex aerofoils improve stability at the cost of aerodynamic performance. However, limited research has focused on combining these two configurations into a hybrid form to achieve an optimal balance suited for modern UAV applications, particularly under varying flight conditions.

Most existing studies emphasize either aerodynamic performance or control stability independently, neglecting the interdependence between lift generation, moment coefficients, and trim drag that defines overall UAV efficiency. Additionally, there is a lack of detailed computational and experimental investigations into how the ratio of cambered to reflex sections influences aerodynamic flow characteristics and stability derivatives.

The present research addresses this gap by employing computational fluid dynamics (CFD) analysis to examine the aerodynamic behaviour of a hybrid reflex-camber aerofoil and compare its performance against conventional profiles. The outcomes will provide valuable insights into designing aerodynamically efficient yet inherently stable UAV wings, contributing to improved endurance, smoother control response, and enhanced mission adaptability.

Furthermore, this study aims to bridge the theoretical understanding and practical application of hybrid aerofoil design, offering a scientific foundation for future UAV development and design optimization efforts. The findings will not only enhance existing aerodynamic knowledge but also serve as a reference for engineers seeking to design next-generation UAVs with improved stability and control characteristic

CHAPTER 2

LITERATURE SURVEY

1. “*Computational fluid dynamics of reflex airfoil*”

Vikas Kumar, Jag Sahil Saini, and Rajkamal Singh

In the study “*Computational Fluid Dynamics of Reflex Airfoil*”, Vikas Kumar, Jag Sahil Saini, and Rajkamal Singh investigated the aerodynamic characteristics of reflex airfoils using computational fluid dynamics techniques. The study focused on analyzing the lift and drag characteristics of reflex airfoil geometries and understanding how the upward curvature at the trailing edge influences aerodynamic performance. Reflex airfoils are characterized by an upward bending of the trailing edge, which produces a positive pitching moment that helps counteract the nose-down moment generated by conventional cambered airfoils. This property makes reflex airfoils particularly suitable for tailless aircraft and flying-wing configurations where horizontal tail surfaces are absent. The researchers created the airfoil geometry using CAD software and performed CFD simulations in ANSYS to evaluate the aerodynamic performance under different flow conditions. The results demonstrated that reflex airfoils provide improved pitching moment characteristics while maintaining acceptable lift and drag performance. The study highlights the importance of reflex airfoil geometry in achieving aerodynamic stability for tailless aircraft configurations.

2. “*Design Characteristics of an Airfoil for Flying/Tailless Wings*”

Sonia Chalia (2017)

In the paper “*Design Characteristics of an Airfoil for Flying/Tailless Wings: A Study*”, Sonia Chalia examined the aerodynamic requirements for airfoils used in flying-wing and tailless aircraft configurations. The study emphasizes that one of the major challenges in flying-wing aircraft design is achieving longitudinal stability without the presence of a conventional horizontal tail. Flying-wing aircraft integrate the fuselage, wings, and propulsion system into a single lifting surface, which significantly reduces aerodynamic drag and improves fuel

efficiency. However, this configuration requires careful airfoil selection to ensure stability and control during flight. The study explains that reflex camber airfoils are commonly used in tailless aircraft because they generate a near-zero or slightly positive pitching moment about the aerodynamic center. By adjusting the camber distribution and trailing edge geometry, the pitching moment characteristics of the airfoil can be controlled to improve stability. The research also discusses historical examples of flying-wing aircraft such as the Northrop flying-wing designs, which demonstrate the practical application of reflex airfoils in real aircraft systems. The findings of the study emphasize that proper airfoil design plays a critical role in ensuring the aerodynamic stability and performance of flying-wing UAVs and tailless aircraft.

3. “Hybrid Airfoil Design for Wind Tunnel Testing”

Gustavo E. C. Fujiwara

In the study “*A Hybrid Airfoil Design Method for Icing Wind Tunnel Tests*”, Gustavo E. C. Fujiwara and co-authors investigated the development of hybrid airfoil geometries for large-scale icing wind tunnel experiments. The primary challenge addressed in this work was the difficulty of testing full-scale aircraft wings in icing wind tunnels due to limitations in tunnel size and scaling constraints. The authors proposed the use of hybrid airfoil geometries that combine the leading-edge characteristics of the full-scale airfoil with a modified aft section. This approach allows the hybrid airfoil to replicate the aerodynamic and icing characteristics of the full-scale airfoil while maintaining compatibility with the smaller test models used in wind tunnel experiments. By carefully modifying the aft geometry of the airfoil, the researchers were able to match key aerodynamic parameters such as pressure distribution, lift characteristics, and pitching moment behavior. The study also analyzed the influence of parameters such as zero-angle pitching moment coefficient, pressure coefficient distribution, and aerodynamic circulation. The results indicated that hybrid airfoils can effectively reproduce the aerodynamic flow characteristics of full-scale airfoils even when the model geometry is altered for experimental constraints. This methodology provides an efficient way to perform aerodynamic and icing tests without requiring full-scale models. The findings of this research demonstrate the potential of hybrid airfoil design techniques in

replicating desired aerodynamic characteristics while accommodating specific experimental or operational constraints. The concept of combining different airfoil geometries to achieve targeted aerodynamic behavior provides valuable insight into the development of optimized airfoil configurations for specialized applications.

4. “*Rationale for Choosing the Airfoil of a UAV Wing Using a Dynamic Ground Effect Principle*”

Andrii Dreus and Olena Kravets

In the paper “*Rationale for Choosing the Airfoil of a UAV Wing Using a Dynamic Ground Effect Principle*”, Andrii Dreus and Olena Kravets investigated the aerodynamic performance of different airfoil shapes for unmanned aerial vehicles operating near the ground. The study focused on vehicles that utilize the ground effect phenomenon, where the aerodynamic lift increases as the aircraft approaches the ground surface. The authors examined several airfoil profiles commonly used in UAV and wing-in-ground (WIG) craft applications, including Clark YH-12, NACA M6, USA-35B, and TsAGI-721. Computational Fluid Dynamics (CFD) modeling was used to analyze the pressure and velocity fields around the airfoils and to determine how the aerodynamic characteristics change as the distance between the wing and ground surface varies. The results showed that airfoil geometry plays a significant role in determining the aerodynamic efficiency and lift enhancement in ground-effect flight. The study found that the aerodynamic quality of the airfoils improved significantly when operating close to the ground, particularly at angles of attack between 4° and 6° . Among the airfoils analyzed, the USA-35B airfoil demonstrated the greatest improvement in aerodynamic efficiency when approaching the ground surface. The authors concluded that the selection of airfoil geometry is a critical factor in UAV wing design, particularly for applications involving ground-effect flight. Their work highlights the importance of evaluating aerodynamic performance through computational simulations in order to determine the most suitable airfoil configuration for specific UAV operating conditions.

5. “*Conceptual Design of an Unmanned Fixed-Wing Aerial Vehicle Based on Alternative Energy*”

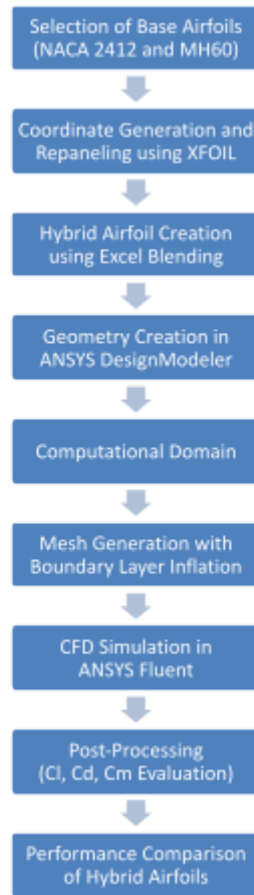
Alan G. Escobar-Ruiz

In the research article “*Conceptual Design of an Unmanned Fixed-Wing Aerial Vehicle Based on Alternative Energy*”, Alan G. Escobar-Ruiz and co-authors presented a conceptual design methodology for developing an unmanned aerial vehicle powered by alternative energy sources such as solar power. The study focuses on improving the endurance and aerodynamic efficiency of UAVs through optimized aerodynamic design. The authors highlighted that aerodynamic efficiency is a crucial factor in UAV performance, especially for vehicles designed for long-endurance missions. The conceptual design process involved aerodynamic analysis of various UAV components including the wing, fuselage, and empennage. Computational Fluid Dynamics (CFD) simulations were used to estimate aerodynamic coefficients and evaluate the lift-to-drag performance of the proposed UAV configuration. The results demonstrated that careful aerodynamic design can significantly improve the endurance and efficiency of UAV systems. The study emphasizes that the selection of suitable airfoil geometry and aerodynamic optimization of the wing structure are essential for achieving improved flight performance and energy efficiency in UAV designs.

CHAPTER 3

METHODOLOGY

This chapter describes the methodology adopted for the design and aerodynamic analysis of hybrid airfoils formed by blending cambered and reflex airfoil profiles. The study focuses on evaluating the aerodynamic performance of these hybrid airfoils using computational fluid dynamics (CFD). The methodology includes the generation of airfoil coordinates, creation of hybrid airfoil geometries, development of the computational domain, mesh generation, turbulence model selection, and application of appropriate boundary conditions. The primary objective of this methodology is to analyze how combining conventional cambered airfoil characteristics with reflex airfoil features influences aerodynamic behaviour, particularly with respect to lift, drag, and pitching moment. All simulations in this study were conducted at an angle of attack of 4° , which represents a typical operating condition for many small UAVs.



3.0.1 Methodology flowchart of hybrid airfoil CFD analysis

3.1 COORDINATE CREATION

The first stage of the methodology involved obtaining and preparing the coordinate data for the base airfoils used in the hybrid airfoil design. Two different airfoils were selected for this study: a conventional cambered airfoil and a reflex airfoil. The cambered airfoil chosen was the NACA 2412 airfoil, which is widely used in aerodynamic studies due to its reliable lift characteristics and well-documented aerodynamic behaviour. The reflex airfoil selected was the MH 60 airfoil, which is known for its reflexed trailing edge that helps reduce the negative pitching moment and improve longitudinal stability, making it suitable for tailless aircraft and UAV configurations.

The coordinate data for both airfoils were obtained from publicly available airfoil databases. However, the coordinate datasets initially contained different numbers of coordinate points along the airfoil surfaces. Since the generation of hybrid airfoils requires corresponding coordinate points between the two airfoils, it was necessary to standardize the number of coordinates before

blending them.

To achieve this, the airfoil geometries were processed using XFOIL. Using the paneling and repaneling functions available in XFOIL, both airfoils were regenerated so that an identical number of coordinate points were distributed along the airfoil surface. In this study, the airfoils were repaneled to generate **160 coordinate points**, ensuring that the upper and lower surface geometry of both airfoils had matching coordinate distributions. This step ensured that each coordinate point from one airfoil corresponded directly to a point on the other airfoil, which is essential for the hybridization process.

Once the coordinates were standardized, the datasets were imported into Microsoft Excel for further processing. In Excel, the hybrid airfoil geometries were generated by mathematically blending the corresponding coordinate points of the two airfoils using different blending ratios. By combining the x and y coordinate values of the two airfoils in varying proportions, several hybrid airfoil profiles were created. These hybrid airfoils were designed to combine the lift characteristics of the cambered airfoil with the improved pitching moment behaviour of the reflex airfoil.

1	1	1.000000	0.001260	0.000000
1	2	0.993813	0.002535	0.000000
1	3	0.983041	0.004730	0.000000
1	4	0.970492	0.007245	0.000000
1	5	0.956456	0.010007	0.000000
1	6	0.941384	0.012914	0.000000
1	7	0.925694	0.015877	0.000000
1	8	0.909666	0.018836	0.000000
1	9	0.893463	0.021760	0.000000
1	10	0.877167	0.024633	0.000000
1	11	0.860819	0.027449	0.000000
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1	38	0.417495	0.077316	0.000000
1	39	0.401335	0.077980	0.000000
1	40	0.385293	0.078517	0.000000

3.1.1 First 40 coordinates of NACA 2412 airfoil

1	1	1.000000	0.000000	0.000000
1	2	0.993813	-0.000169	0.000000
1	3	0.983041	-0.000183	0.000000
1	4	0.970492	0.000166	0.000000
1	5	0.956456	0.000811	0.000000
1	6	0.941384	0.001679	0.000000
1	7	0.925694	0.002722	0.000000
1	8	0.909666	0.003924	0.000000
1	9	0.893463	0.005272	0.000000
1	10	0.877167	0.006752	0.000000
1	11	0.860819	0.008350	0.000000
1	12	0.844439	0.010052	0.000000
1	13	0.828038	0.011841	0.000000
1	14	0.811621	0.013705	0.000000
1	15	0.795192	0.015632	0.000000
1	16	0.778752	0.017612	0.000000
1	17	0.762303	0.019638	0.000000
1	18	0.745847	0.021701	0.000000
1	19	0.729385	0.023794	0.000000
1	20	0.712918	0.025909	0.000000
1	21	0.696448	0.028038	0.000000
1	22	0.679976	0.030173	0.000000
1	23	0.663504	0.032305	0.000000
1	24	0.647032	0.034422	0.000000
1	25	0.630563	0.036513	0.000000
1	26	0.614097	0.038570	0.000000
1	27	0.597636	0.040585	0.000000
1	28	0.581182	0.042554	0.000000
1	29	0.564737	0.044472	0.000000
1	30	0.548301	0.046335	0.000000
1	31	0.531878	0.048137	0.000000
1	32	0.515468	0.049876	0.000000
1	33	0.499074	0.051548	0.000000
1	34	0.482700	0.053149	0.000000
1	35	0.466347	0.054679	0.000000
1	36	0.450022	0.056134	0.000000
1	37	0.433733	0.057513	0.000000
1	38	0.417495	0.058815	0.000000
1	39	0.401335	0.060040	0.000000
1	40	0.385293	0.061189	0.000000

3.1.2 First 40 coordinates of MH 60 airfoil

x	y_2412	y_mh60	y_hybrid 90:10	y_hybrid 80:20	y_hybrid 70:30	y_hybrid 60:40	y_hybrid 50:50	y_hybrid 40:60	y_hybrid 30:70	y_hybrid 20:80	y_hybrid 10:90
1	0.00126	0	0.001134	0.001008	0.000882	0.000756	0.00063	0.000504	0.000378	0.000252	0.000126
0.993813	0.002535	-0.000169	0.0022646	0.0019942	0.0017238	0.0014534	0.001183	0.0009126	0.0006422	0.0003718	0.0001014
0.983041	0.00473	-0.000183	0.0042387	0.0037474	0.0032561	0.0027648	0.0022735	0.0017822	0.0012909	0.0007996	0.0003083
0.970492	0.007245	0.000166	0.0065371	0.0058292	0.0051213	0.0044134	0.0037055	0.0029976	0.0022897	0.0015818	0.0008739
0.956456	0.010007	0.000811	0.0090874	0.0081678	0.0072482	0.0063286	0.005409	0.0044894	0.0035698	0.0026502	0.0017306
0.941384	0.012914	0.001679	0.0117905	0.010667	0.0095435	0.00842	0.0072965	0.006173	0.0050495	0.003926	0.0028025
0.925694	0.015877	0.002722	0.0145615	0.013246	0.0119305	0.010615	0.0092995	0.007984	0.0066685	0.005353	0.0040375
0.909666	0.018836	0.003924	0.0173448	0.0158536	0.0143624	0.0128712	0.01138	0.0098888	0.0083976	0.0069064	0.0054152
0.893463	0.02176	0.005272	0.0201112	0.0184624	0.0168136	0.0151648	0.013516	0.0118672	0.0102184	0.0085696	0.0069208
0.877167	0.024633	0.006752	0.0228449	0.0210568	0.0192687	0.0174806	0.0156925	0.0139044	0.0121163	0.0103282	0.0085401
0.860819	0.027449	0.00835	0.0255391	0.0236292	0.0217193	0.0198094	0.0178995	0.0159896	0.0140797	0.0121698	0.0102599
0.844439	0.030203	0.010052	0.0281879	0.0261728	0.0241577	0.0221426	0.0201275	0.0181124	0.0160973	0.0140822	0.0120671
0.828038	0.032894	0.011841	0.0307887	0.0286834	0.0265781	0.0244728	0.0223675	0.0202622	0.0181569	0.0160516	0.0139463
0.811621	0.035522	0.013705	0.0333403	0.0311586	0.0289769	0.0267952	0.0246135	0.0224318	0.0202501	0.0180684	0.0158867
0.795192	0.038086	0.015632	0.0358406	0.0335952	0.0313498	0.0291044	0.026859	0.0246136	0.0223682	0.0201228	0.0178774
0.778752	0.040585	0.017612	0.0382877	0.0359904	0.0336931	0.0313958	0.0290985	0.0268012	0.0245039	0.0222066	0.0199093
0.762303	0.04302	0.019638	0.0406818	0.0383436	0.0360054	0.0336672	0.031329	0.0289908	0.0266526	0.0243144	0.0219762
0.745847	0.045389	0.021701	0.0430202	0.0406514	0.0382826	0.0359138	0.033545	0.0311762	0.0288074	0.0264386	0.0240698
0.729385	0.047693	0.023794	0.0453031	0.0429132	0.0405233	0.0381334	0.0357435	0.0333536	0.0309637	0.0285738	0.0261839
0.712918	0.049931	0.025909	0.0475288	0.0451266	0.0427244	0.0403222	0.03792	0.0355178	0.0331156	0.0307134	0.0283112
0.696448	0.052101	0.028038	0.0496947	0.0472884	0.0448821	0.0424758	0.0400695	0.0376632	0.0352569	0.0328506	0.0304443
0.679976	0.054204	0.030173	0.0518009	0.0493978	0.0469947	0.0445916	0.0421885	0.0397854	0.0373823	0.0349792	0.0325761
0.663504	0.056238	0.032305	0.0538447	0.0514514	0.0490581	0.0466648	0.0442715	0.0418782	0.0394849	0.0370916	0.0346983
0.647032	0.058201	0.034422	0.0558231	0.0534452	0.0510673	0.0486894	0.0463115	0.0439336	0.0415557	0.0391778	0.0367999
0.630563	0.060093	0.036513	0.057735	0.055377	0.053019	0.050661	0.048303	0.045945	0.043587	0.041229	0.038871

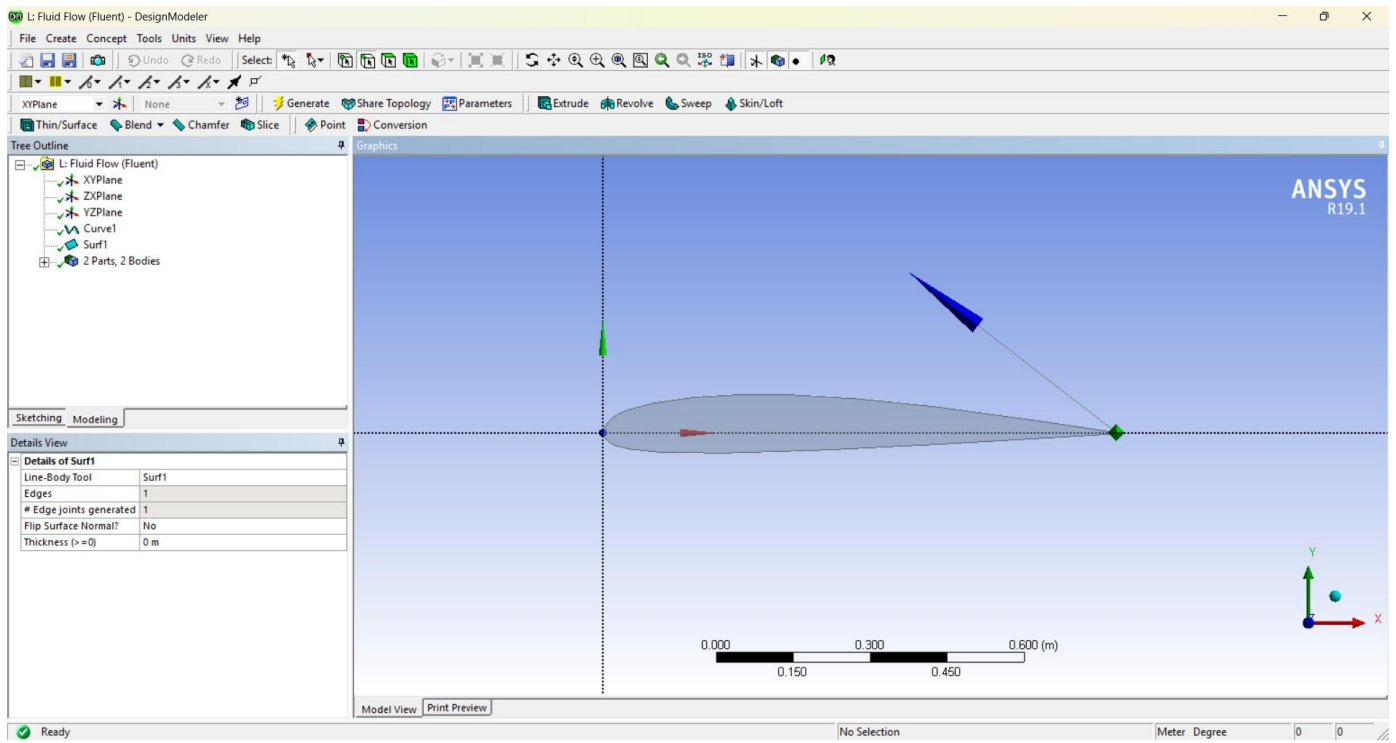
3.1.3 Coordinates of hybrid airfoils of different ratios

3.2 GEOMETRY CREATION

The generated airfoil coordinate data was imported into ANSYS Design Modeler to create the airfoil geometry required for the CFD analysis. The coordinate points obtained from the hybrid airfoil generation process were first arranged in a suitable format and then imported into the geometry environment. Using these coordinate points, a 3-D curve was created in Design Modeler to represent the airfoil profile accurately. The curve connected all the coordinate points sequentially, thereby forming the complete airfoil shape based on the generated coordinate data.

Once the airfoil curve was generated, the geometry was carefully checked to ensure that the leading edge and trailing edge were properly defined and that the curve formed a continuous airfoil profile without any gaps. This step is important because an accurate geometric representation of the airfoil is essential for obtaining reliable CFD simulation results. After verifying the geometry, separate airfoil profiles were created for each hybrid airfoil configuration developed during the coordinate blending process.

The airfoil geometry was then positioned inside the computational domain for aerodynamic analysis. To simulate realistic operating conditions, the airfoil was oriented at an angle of attack of 4° relative to the incoming airflow. This orientation allowed the CFD solver to evaluate the aerodynamic characteristics of the airfoil, including lift, drag, and pitching moment, under the specified operating condition. Proper alignment and positioning of the airfoil within the domain ensured that the subsequent meshing and simulation stages could be carried out accurately.

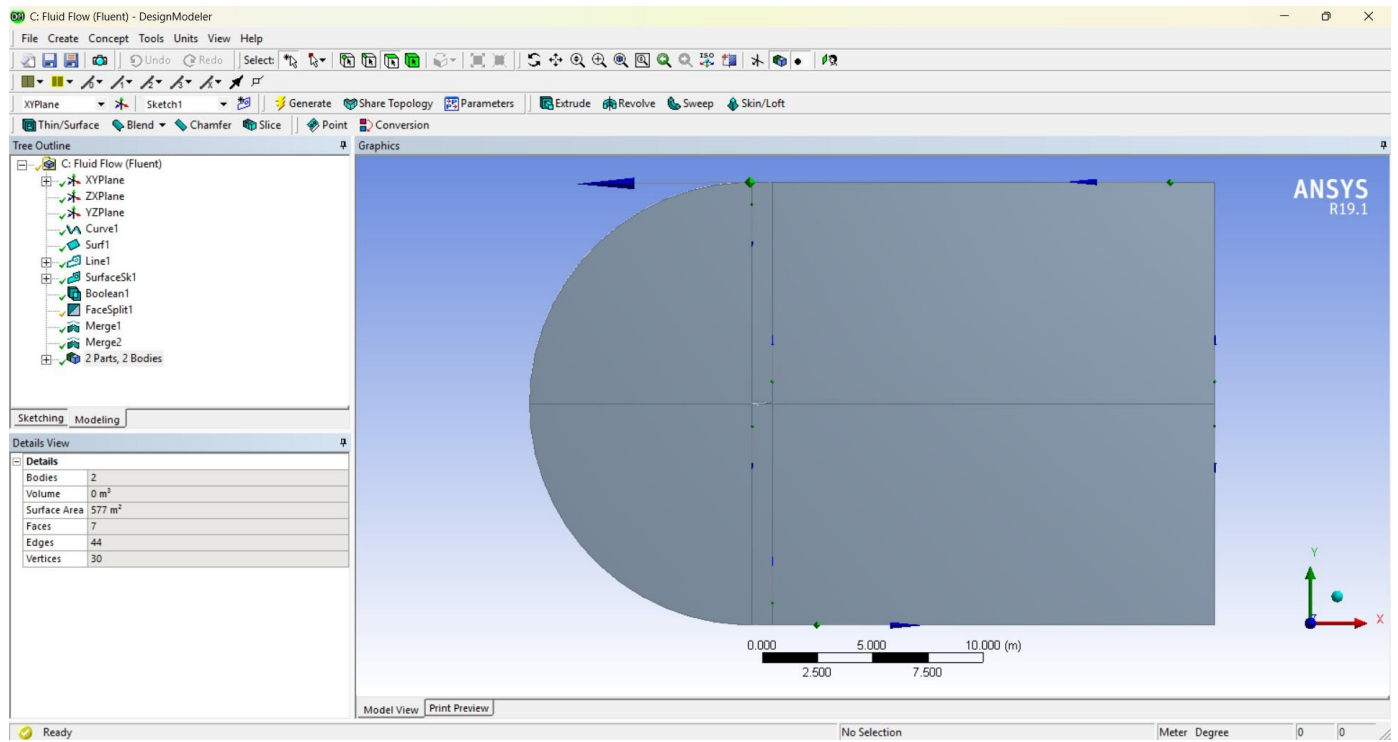


3.2.1 Airfoil geometry created in ANSYS

3.3 COMPUTATIONAL DOMAIN

To perform the CFD analysis, a two-dimensional C-type computational domain was created around the airfoil geometry. The C-domain configuration is commonly used for external aerodynamic simulations of airfoils because it allows smooth airflow development around the leading edge while minimizing numerical errors near the outlet boundary. In this domain type, the inlet boundary surrounds the leading edge in a curved “C” shape, which helps capture the flow behaviour around the airfoil more accurately. The size of the computational domain was carefully selected to minimize the influence of the boundaries on the flow field around the airfoil. All domain dimensions were defined relative to the airfoil chord length (c) to maintain geometric scalability. The inlet boundary was positioned approximately 10 chord lengths upstream of the airfoil, allowing the incoming airflow to develop before interacting with the airfoil surface. The outlet boundary was placed about 15 chord lengths downstream to ensure that the wake region behind the airfoil could develop fully without interference from the outlet boundary condition.

Similarly, the upper and lower boundaries of the domain were placed approximately 10 chord lengths away from the airfoil. Positioning these boundaries far from the airfoil ensures that the free-stream flow conditions are maintained and prevents artificial blockage effects that could influence the simulation results. The airfoil was located approximately at the center of the domain in the vertical direction to maintain symmetric flow conditions relative to the far-field boundaries. The use of a sufficiently large C-type domain helps ensure that the airflow around the airfoil behaves similarly to real free-stream conditions, thereby improving the accuracy of the predicted aerodynamic parameters such as lift coefficient, drag coefficient, and pitching moment coefficient. After creating the airfoil geometry and the surrounding computational domain, the domain was divided into multiple regions using the Face Split operation available in ANSYS Design Modeler. The face split operation was used to subdivide the domain into several smaller faces in order to provide better control over the mesh distribution during the meshing process. This allowed local mesh refinement to be applied near the airfoil surface where strong velocity and pressure gradients occur, while maintaining a relatively coarser mesh in regions farther away from the airfoil. The use of face splitting therefore helped improve mesh quality and ensured more accurate CFD simulation results.



3.3.1 Computational domain created around the airfoil.

3.4 MESHING

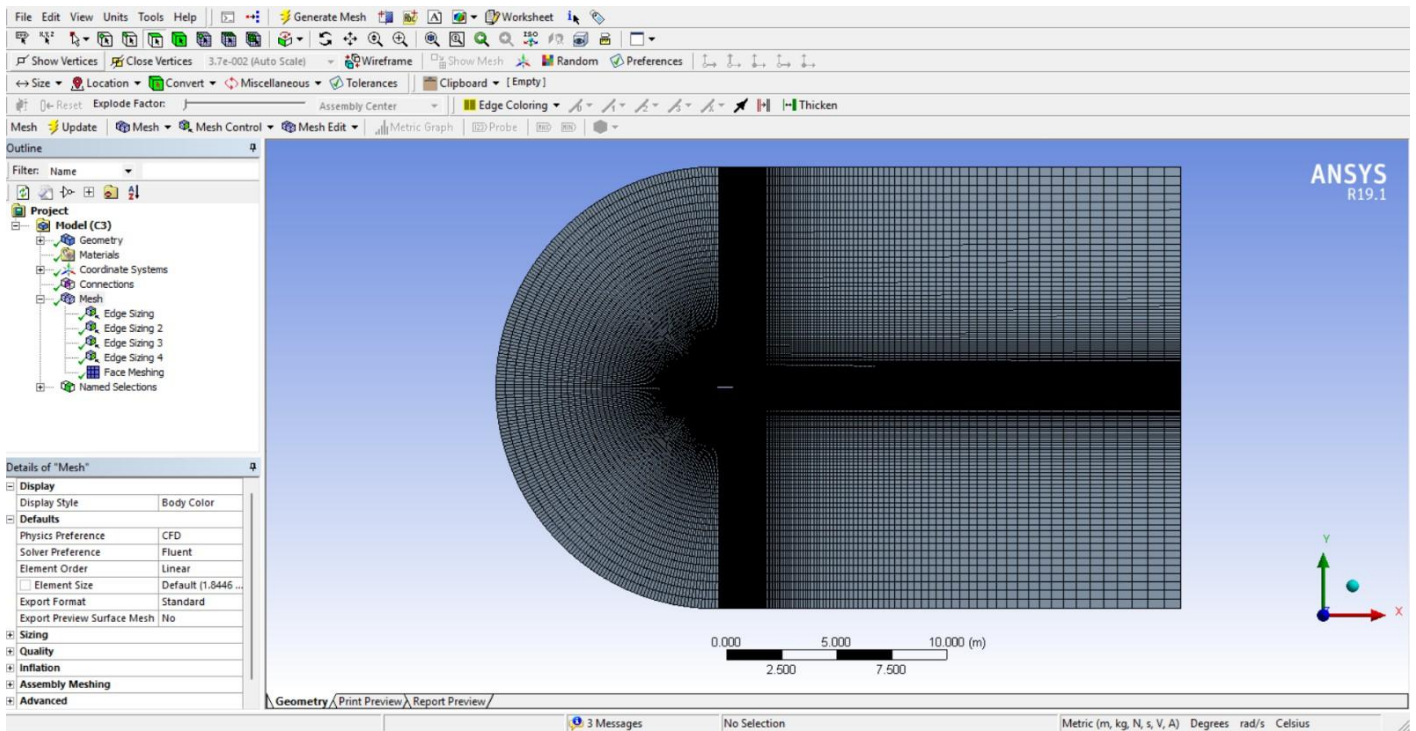
The computational domain was discretized into smaller elements using the meshing module available in ANSYS Meshing. Mesh generation is a critical step in computational fluid dynamics analysis because the accuracy and stability of the numerical solution depend strongly on the quality of the mesh. A suitable mesh must be fine enough to capture the important flow features around the airfoil while also maintaining reasonable computational cost.

In this study, a structured or semi-structured mesh was generated around the airfoil geometry to capture the aerodynamic flow behaviour accurately. Higher mesh refinement was applied in the vicinity of the airfoil surface where strong velocity and pressure gradients occur. This refinement is essential for accurately resolving the boundary layer region and predicting aerodynamic forces such as lift, drag, and pitching moment.

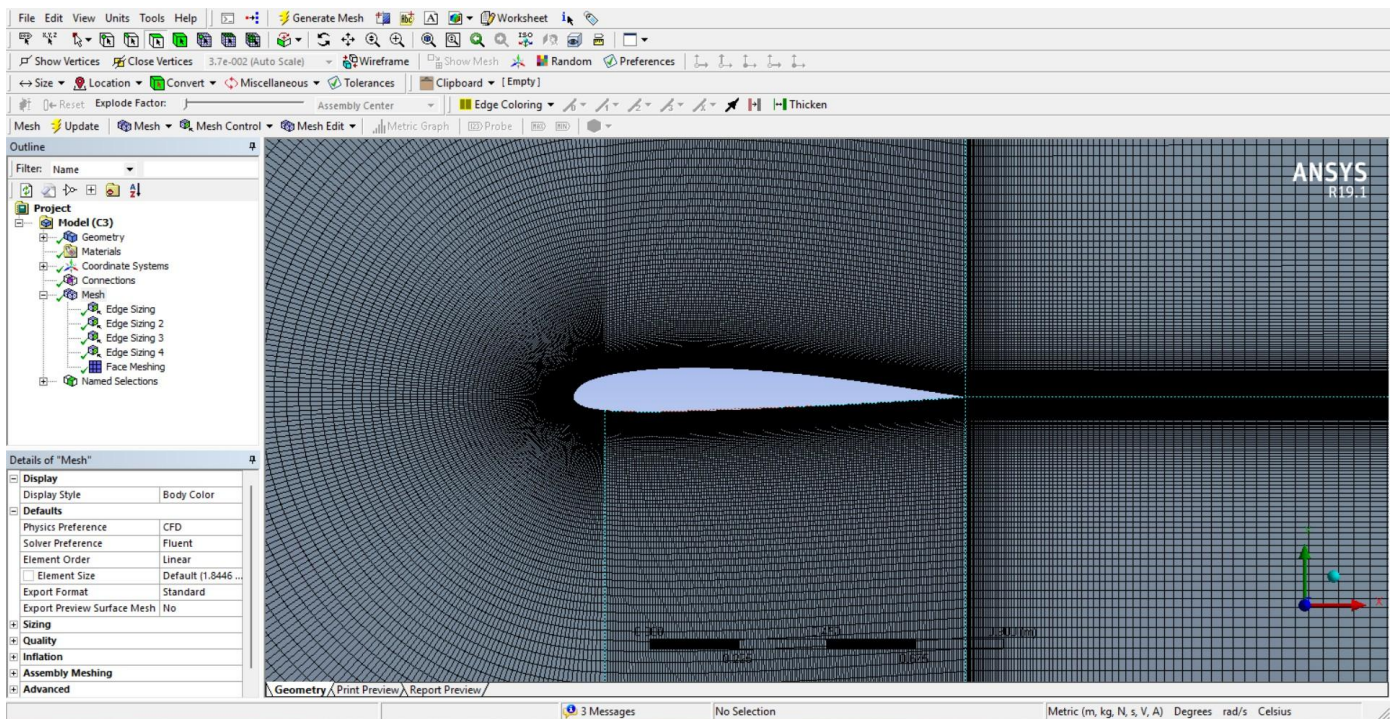
To properly capture the boundary layer flow near the airfoil wall, inflation layers were applied along the airfoil surface. The near-wall mesh was designed based on the dimensionless wall distance parameter (y^+), which determines the distance of the first mesh node from the wall in turbulent flow simulations. For the selected flow conditions, including the inlet velocity and

Reynolds number, the calculated y^+ value was approximately 1.7564×10^{-5} m. Based on this calculation, the first layer thickness of the inflation mesh was set to 8.525×10^{-6} m in order to accurately resolve the viscous sublayer region of the boundary layer.

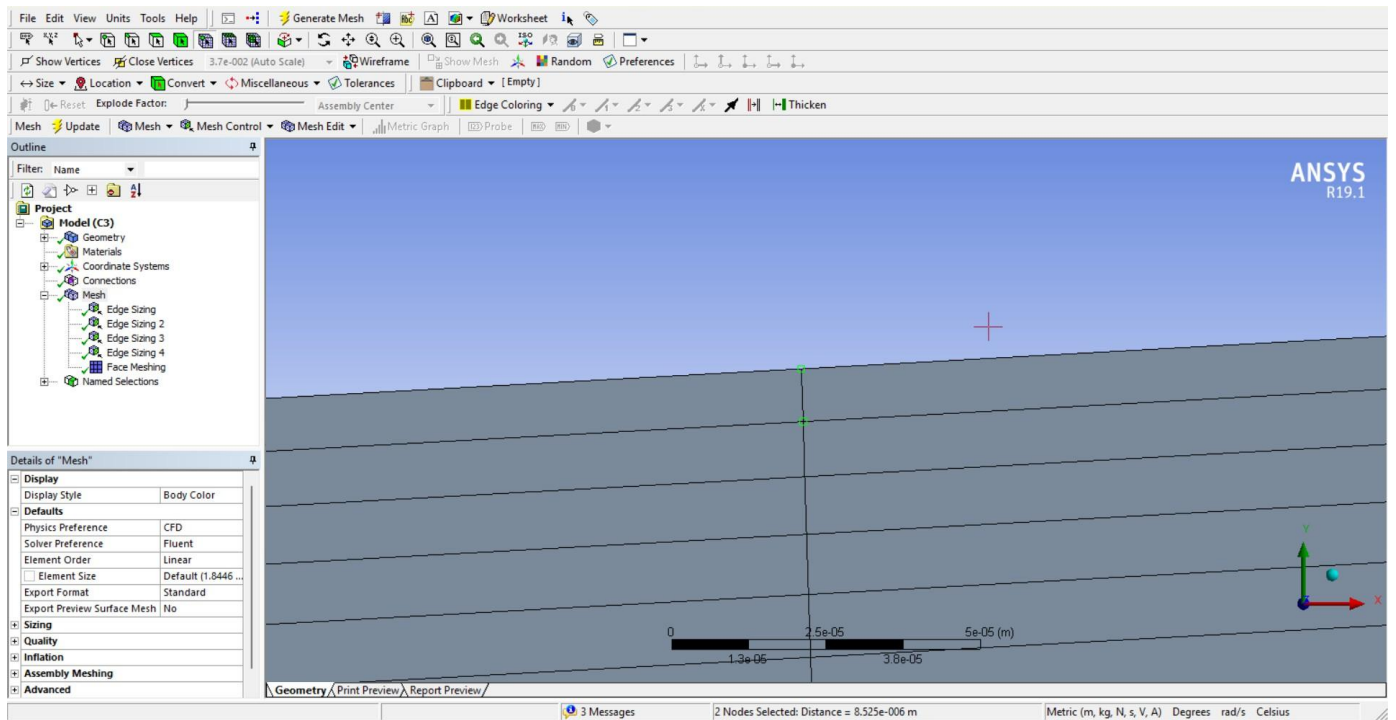
Additional mesh control techniques were applied to improve mesh quality and ensure accurate representation of the flow field. Edge sizing was used along selected edges of the computational domain and airfoil geometry to control the distribution and density of mesh elements. Furthermore, biasing was applied along specific edges to gradually vary the element size from finer regions near the airfoil surface to coarser regions farther away from the airfoil. This approach helps maintain high mesh resolution where it is most needed while reducing the total number of elements in the domain. In certain regions of the computational domain, hard mesh behaviour was applied to enforce strict mesh sizing constraints. This ensured that the mesh followed the specified element sizes and refinement levels in critical areas of the domain. The combination of edge sizing, biasing, and inflation layers resulted in a mesh that effectively captured the aerodynamic flow features around the airfoil while maintaining acceptable mesh quality parameters.



3.4.1 Mesh around the airfoil



3.4.2 Zoomed view of mesh near airfoil surface



3.4.3 First layer thickness near airfoil surface

3.5 MESH INDEPENDENCE STUDY

A mesh independence study was conducted to ensure that the numerical results obtained from the CFD simulations are reliable and not significantly influenced by the mesh size or element distribution. This step is essential in validating the accuracy of the computational model.

In this study, simulations were performed using different mesh densities by systematically increasing the number of elements in the computational domain. The mesh size was varied from approximately **20,000 elements to 200,000 elements**, while keeping all other simulation parameters constant.

The simulations were carried out at:

- **Angle of Attack:** 4°
- **Free-stream Velocity:** 20 m/s

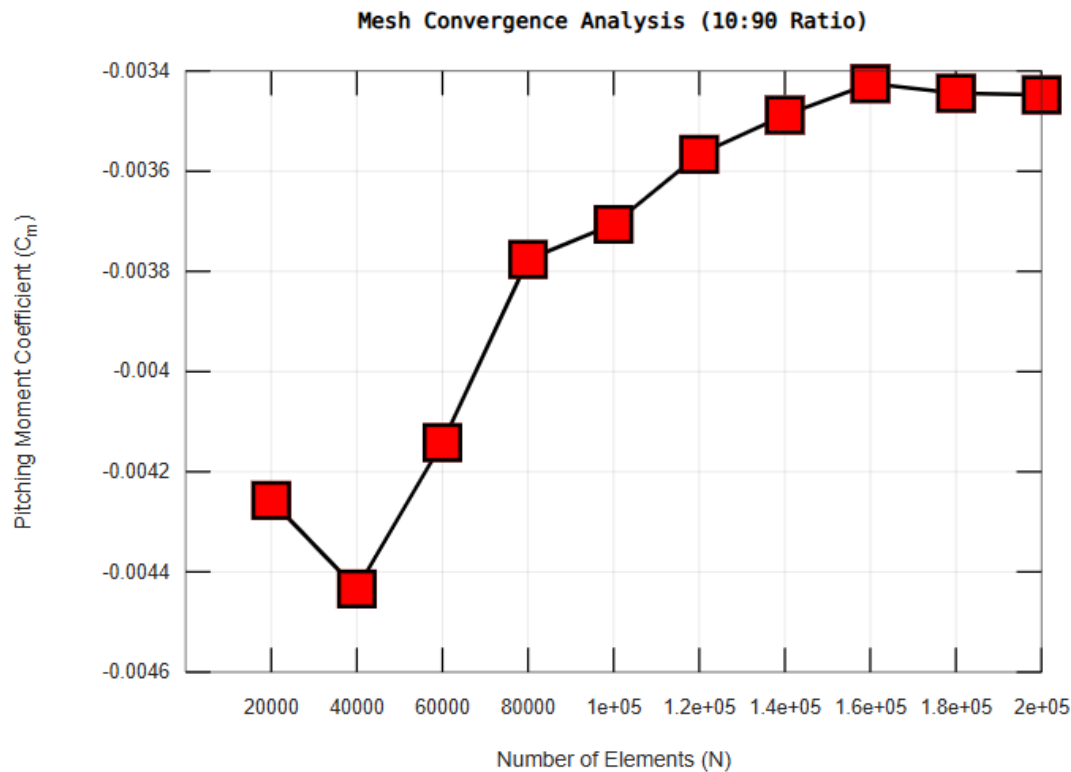
The **pitching moment coefficient (C_m)** was selected as the monitoring parameter because it is highly sensitive to changes in flow resolution and directly related to aerodynamic stability. The relationship between the number of mesh elements and the monitored C_m value is detailed below:

Number of elements	C_m (10:90)
20,000	-0.0042571254
40,000	-0.0044339067
60,000	-0.0041418389
80,000	-0.0037762048
1,00,000	-0.0037062108
1,20,000	-0.0035664092
1,40,000	-0.0034880351
1,60,000	-0.0034255242
1,80,000	-0.003445769
2,00,000	-0.0034477953

3.5.1 Mesh Independence Study

CONVERGENCE BEHAVIOR:

At lower mesh counts (below 80,000 elements), the C_m values exhibited significant fluctuations. A coarse mesh of 20,000 elements yielded a relative error of approximately **9.6%** compared to the final converged value, indicating that the grid was too sparse to capture the boundary layer effects and the reflex curvature physics.



Elements	Cm Value	Error (%)
20000	-0.0042571254	0.00000
40000	-0.0044339067	3.98703
60000	-0.0041418389	7.05165
80000	-0.0037762048	9.68258
100000	-0.0037062108	1.88856
120000	-0.0035664092	3.91995
140000	-0.0034880351	2.24694
160000	-0.0034255242	1.82486
180000	-0.0034445769	0.55312
200000	-0.0034477953	0.09335

3.5.1 Mesh Independence Study

As the element count exceeded 150,000, the variation in aerodynamic coefficients began to diminish. The curve started to plateau, suggesting that the solver was reaching a stable numerical solution.

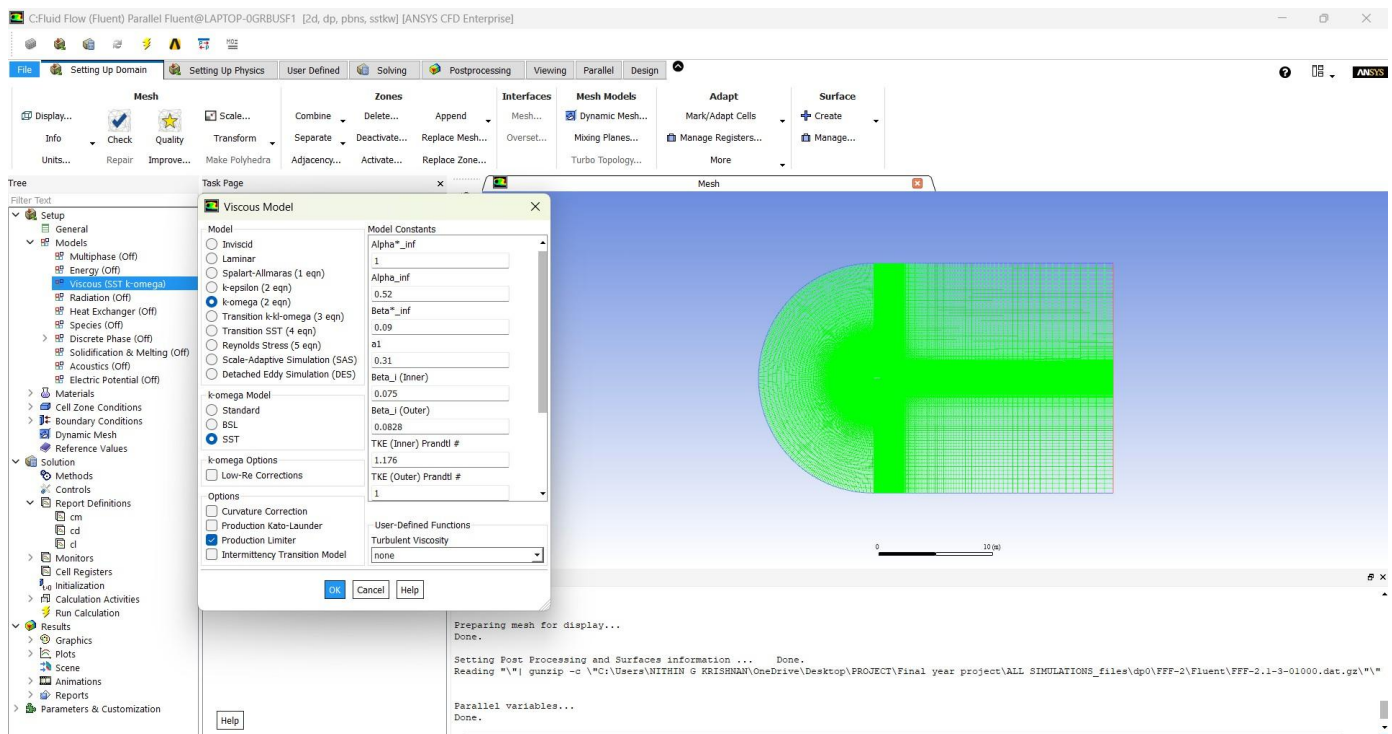
Upon increasing the mesh from **180,000 to 200,000 elements**, the change in the C_m value was found to be only **0.09335%**. Based on these results, a mesh density of approximately 180,000 elements was selected for all subsequent simulations to balance computational efficiency with high numerical accuracy.

3.6 TURBULENCE MODEL

To simulate the aerodynamic behaviour of the airflow around the airfoil, an appropriate turbulence model was selected in the CFD solver. In this study, the $k-\omega$ SST turbulence model was used due to its proven capability in predicting aerodynamic flows around airfoils.

The $k-\omega$ SST model combines the advantages of both the $k-\omega$ and $k-\epsilon$ turbulence models. It provides accurate predictions of boundary layer flow close to the wall while also effectively modelling free-stream turbulence. This hybrid approach makes the $k-\omega$ SST model particularly suitable for external aerodynamic applications such as airfoil analysis.

The simulations were performed using ANSYS Fluent, which solves the governing equations of fluid flow using numerical methods. The turbulence model helps account for the effects of turbulent flow structures that occur around the airfoil at practical operating conditions.



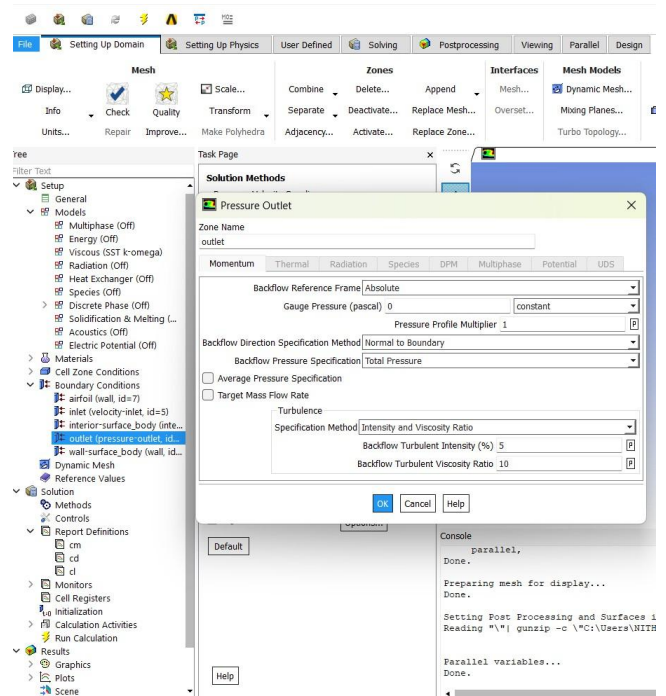
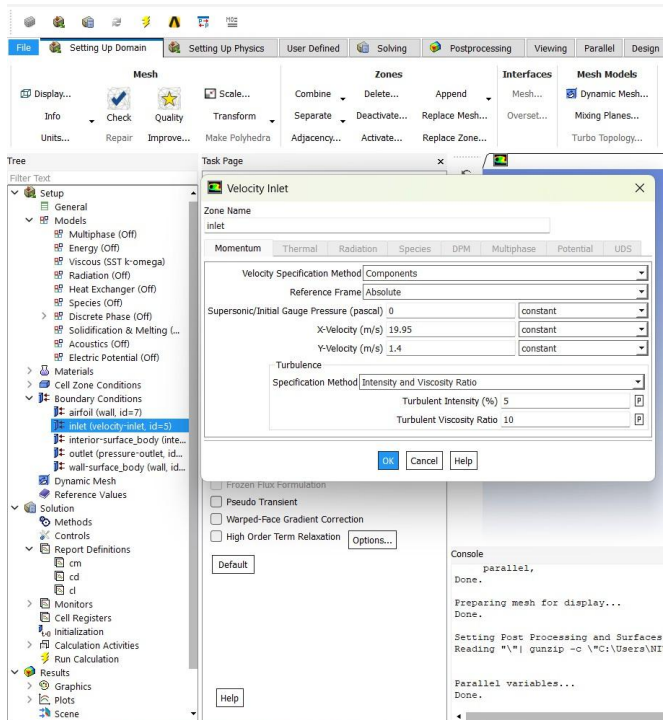
3.6.1 Turbulence model

3.7 SOLVER SETTINGS

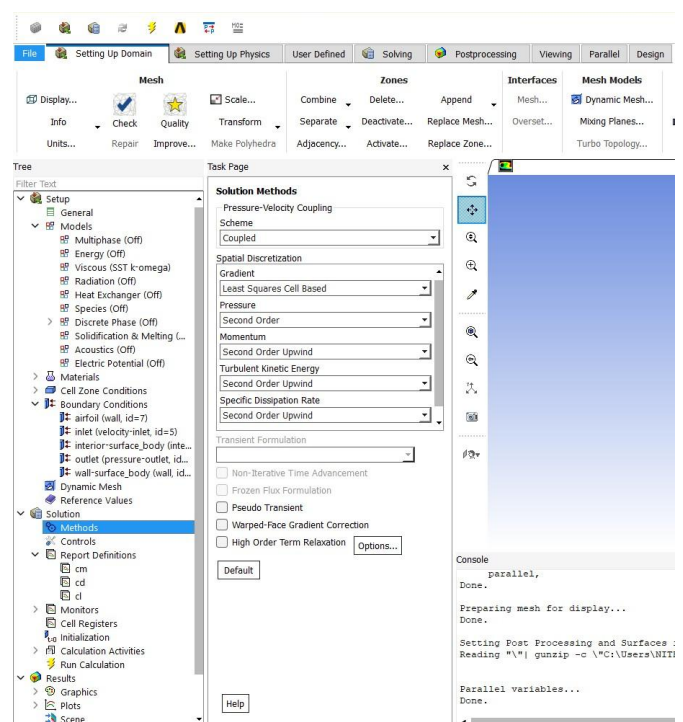
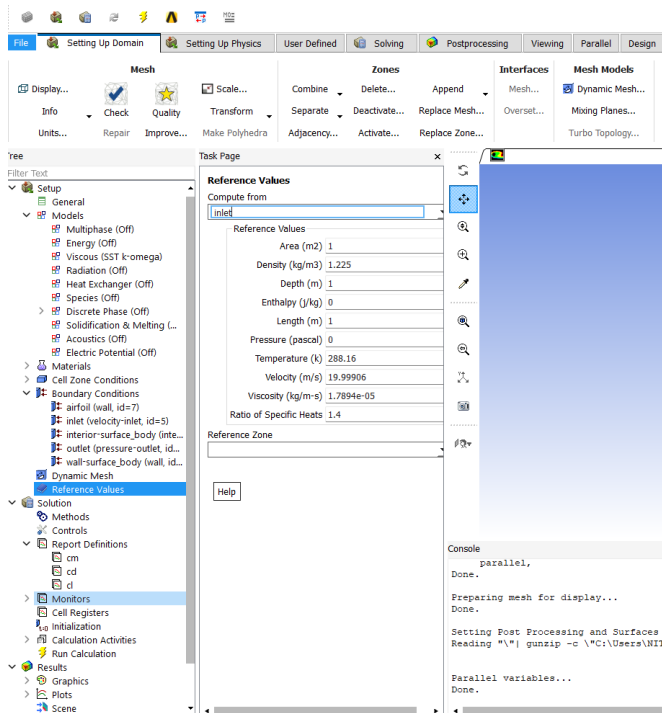
The aerodynamic simulations were carried out using the CFD solver available in ANSYS Fluent. A pressure-based steady-state solver was used for the simulations since the airflow around the airfoil was assumed to be incompressible and steady. Air was defined as the working fluid with its properties set according to standard atmospheric conditions. The $k-\omega$ SST turbulence model was selected for this study due to its ability to accurately predict boundary layer behaviour and flow separation for external aerodynamic applications such as airfoil analysis.

Appropriate boundary conditions were applied to represent the physical flow conditions. A velocity inlet boundary condition was applied at the upstream boundary of the computational domain to define the incoming airflow velocity and velocity of 20m/s was given in components of x and y for 4° AOA ($x=19.95$ m/s and $y=1.40$ m/s). A pressure outlet boundary condition was used at the downstream boundary to allow the airflow to exit the domain without creating artificial pressure build-up. The airfoil surface was defined as a no-slip wall boundary condition, ensuring that the velocity of the fluid at the wall surface is zero. The upper and lower boundaries of the domain were treated as symmetry boundaries to simulate free-stream conditions.

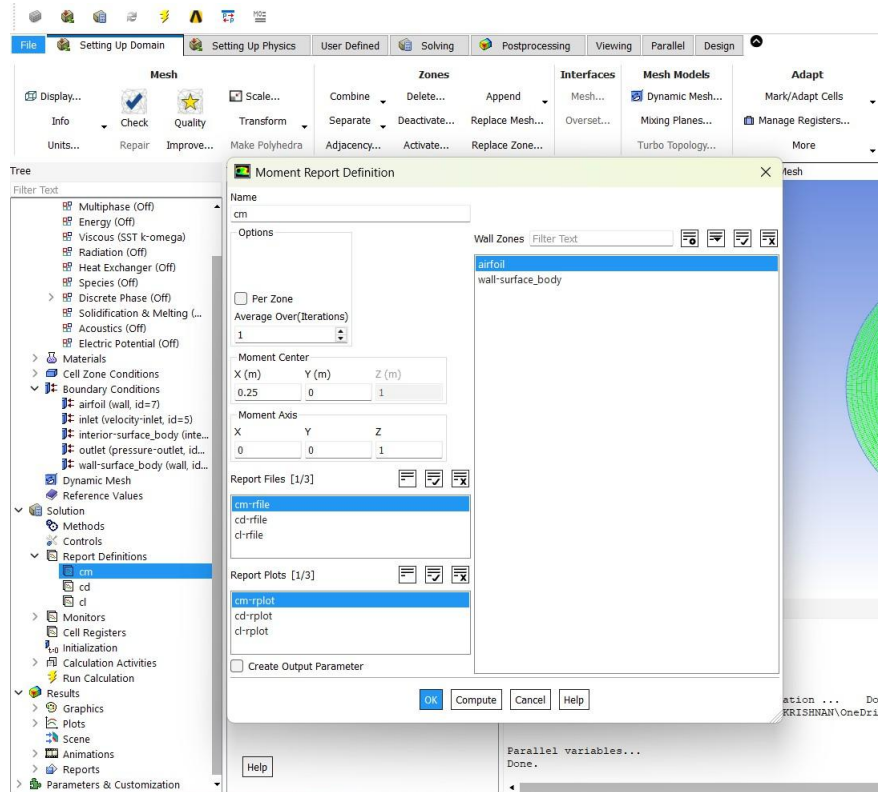
The governing equations were solved using appropriate numerical schemes to ensure solution stability and accuracy. The coupled algorithm was used for pressure–velocity coupling. Second-order discretization schemes were used for the momentum equations and turbulence quantities in order to improve the accuracy of the numerical solution. The solution was iterated until the residuals of the governing equations dropped below 10^{-6} to ensure adequate convergence. During the iteration process, the stability of the solution was monitored to verify that the aerodynamic coefficients reached steady values. To evaluate the aerodynamic performance of the hybrid airfoils, report definitions were created within ANSYS Fluent. These report definitions were used to monitor important aerodynamic coefficients during the simulation process. Specifically, report definitions were created for the lift coefficient (C_l), drag coefficient (C_d), and moment coefficient (C_m) acting on the airfoil surface. These quantities were monitored throughout the iterations to ensure convergence and to obtain the final aerodynamic performance values for each hybrid airfoil configuration. The results obtained from these reports were later used for comparison of the different hybrid airfoil ratios considered in the study.



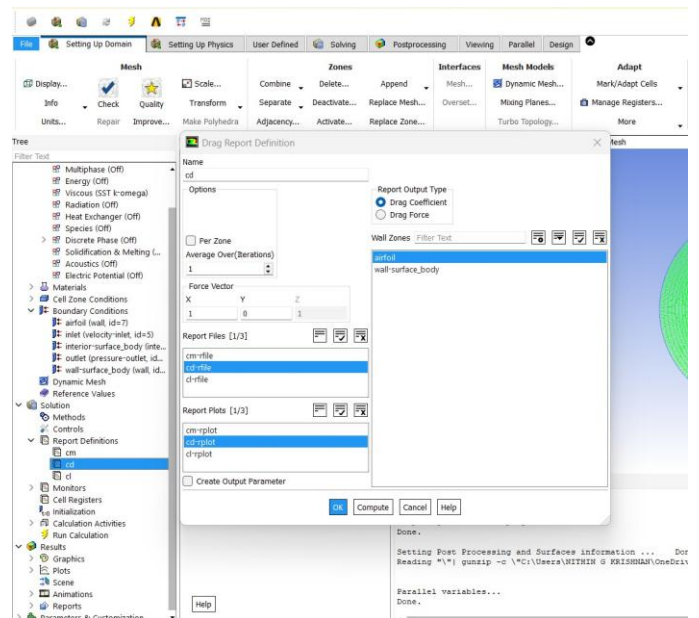
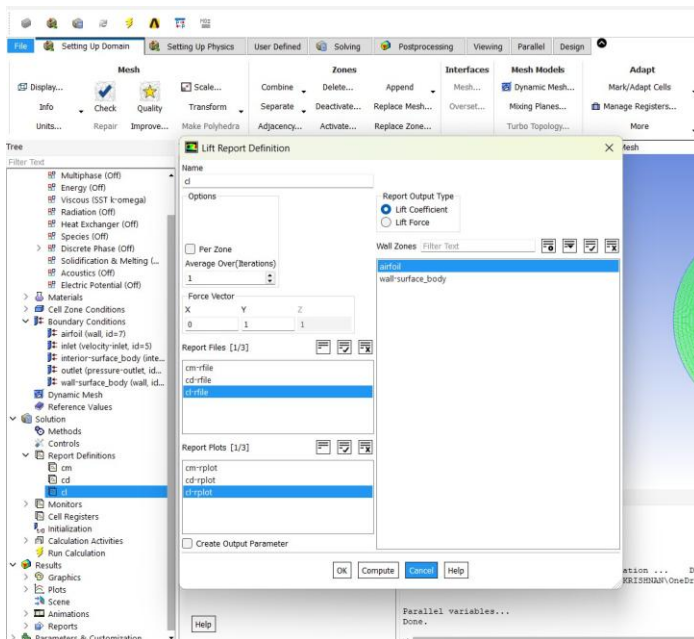
3.7.1 Inlet and outlet boundary conditions



3.7.2 Reference values and Solution methods



3.7.3 Cm report definition



3.7.4 Cl and Cd report definitions

CHAPTER 4

RESULTS AND DISCUSSIONS

This chapter presents the results obtained from the computational fluid dynamics simulations of the hybrid airfoils. The simulations were carried out at an angle of attack of 4° to evaluate the aerodynamic characteristics of the airfoils under a typical operating condition. The aerodynamic performance of the hybrid airfoils was analyzed in terms of lift coefficient, drag coefficient, and pitching moment coefficient. The results obtained from the CFD analysis were then compared to understand how different blending ratios of cambered and reflex airfoils influence aerodynamic behaviour.

4.1 AERODYNAMIC COEFFICIENT ANALYSIS

The aerodynamic performance of the hybrid airfoils was evaluated using the lift coefficient (C_l), drag coefficient (C_d), and pitching moment coefficient (C_m) obtained from the CFD simulations performed in ANSYS Fluent. These coefficients provide insight into the lifting capability, aerodynamic efficiency, and stability characteristics of the airfoil configurations.

RATIO (NACA 2412:MH60)	C_m	C_l	C_d
90:10	0.025600342	0.56881004	0.02777761
80:20	0.022881143	0.56893488	0.027896876
70:30	0.021249638	0.57321102	0.028033101
60:40	0.016030375	0.56370399	0.027687088
50:50	0.011519987	0.5566446	0.027252808
40:60	0.0080092713	0.55321143	0.027072613
30:70	0.0046521049	0.55073782	0.026939179
20:80	0.0014064228	0.54866731	0.02684063
10:90	-0.0034585066	0.53940216	0.026208718

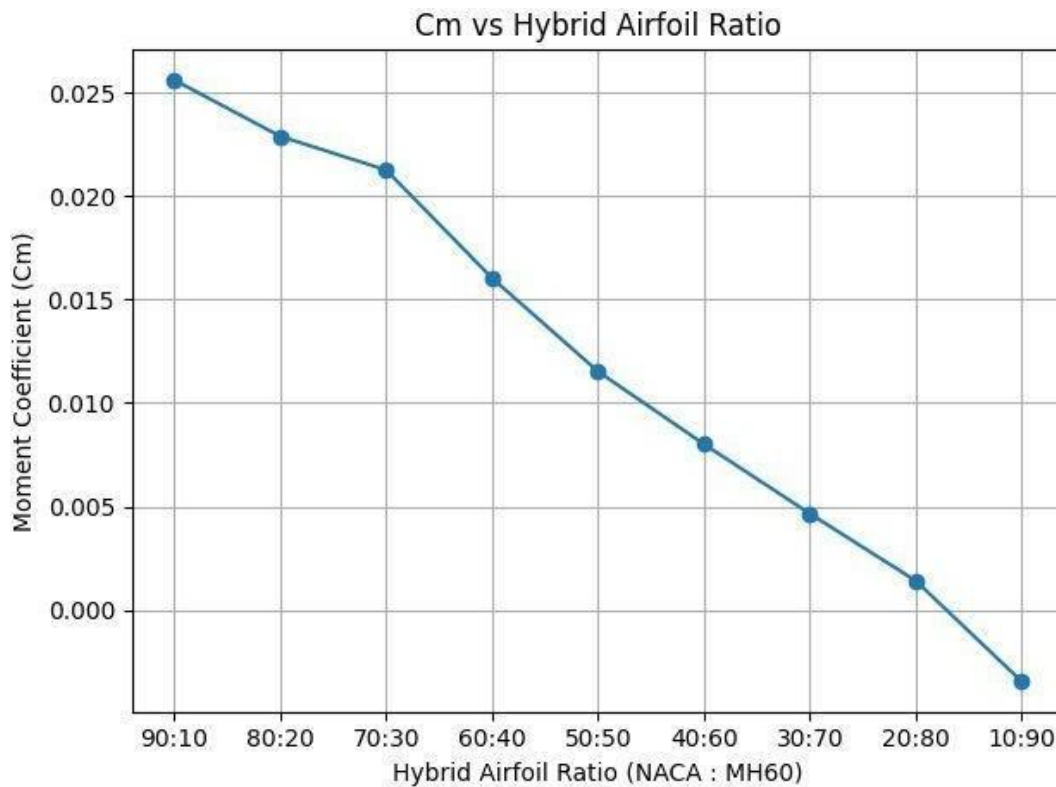
4.1.1 Aerodynamic coefficients of Hybrid airfoils at 4° AOA

Pitching Moment Coefficient (C_m)

The pitching moment coefficient is an important parameter for evaluating the longitudinal stability of an airfoil. Conventional cambered airfoils typically produce a negative pitching moment, which can lead to stability issues in tailless aircraft and UAV configurations.

The results show that as the percentage of the reflex airfoil increases, the pitching moment coefficient gradually decreases and approaches zero. The 10:90 hybrid configuration produced a slightly negative moment coefficient (-0.00346), indicating improved aerodynamic stability compared to configurations with higher camber contribution.

This behaviour is expected because reflex airfoils are designed to reduce or neutralize the pitching moment, making them suitable for applications such as UAV wings and tailless aircraft.



4.1.1 Variation of Moment Coefficient (C_m) with Hybrid Airfoil Ratio

The graph shows the variation of the pitching moment coefficient (C_m) for different hybrid airfoil ratios of NACA and MH60. As the proportion of the reflex airfoil increases, the moment coefficient gradually decreases and approaches zero. This trend indicates that the inclusion of the

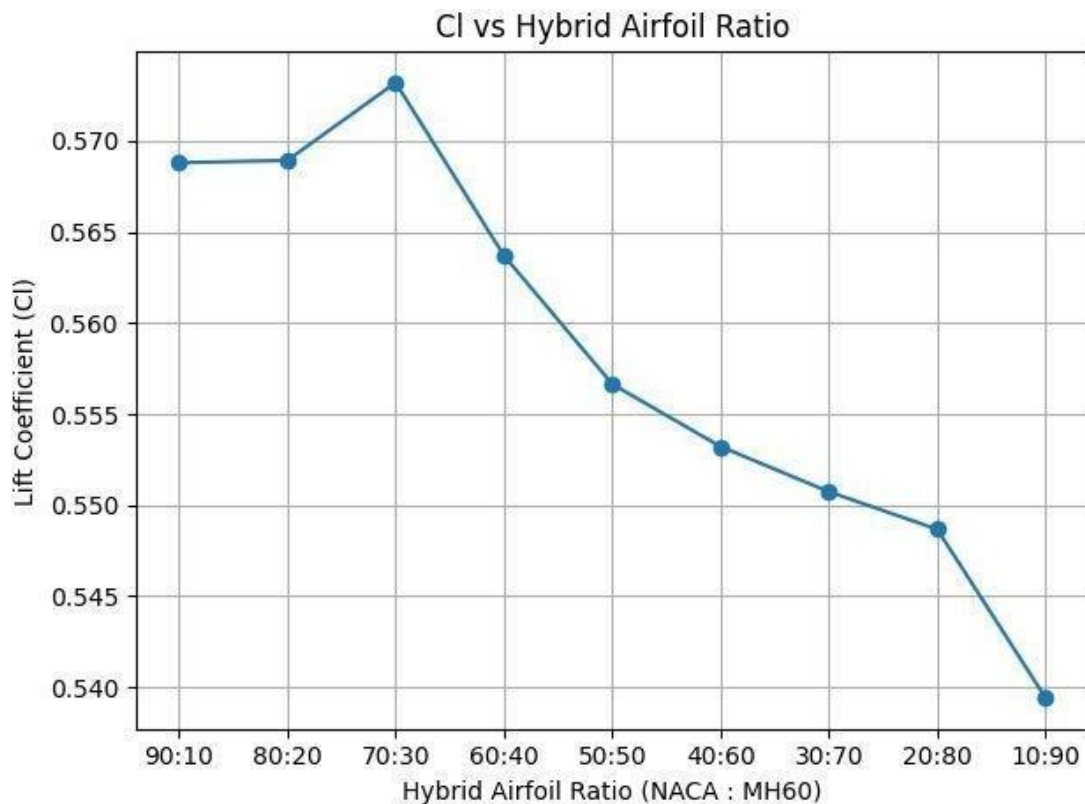
reflex airfoil reduces the pitching moment generated by the cambered airfoil. The lowest moment coefficient is observed for the 10:90 configuration, which produces a slightly negative value. This behaviour confirms that reflex airfoil characteristics improve the longitudinal stability of the airfoil, making such configurations suitable for UAV applications.

Lift Coefficient (Cl)

The lift coefficient represents the ability of the airfoil to generate lift under the given flow conditions. From the simulation results, it can be observed that the lift coefficient slightly increases as the percentage of the cambered airfoil increases in the hybrid configuration.

Among the tested configurations, the 70:30 hybrid airfoil ratio (NACA 2412:MH60) produced the highest lift coefficient of approximately 0.5732, indicating better lifting performance compared to the other configurations. The lift coefficient gradually decreases as the proportion of the reflex airfoil increases, reaching the lowest value of approximately 0.5394 for the 10:90 ratio.

This trend occurs because cambered airfoils generally generate higher lift due to greater curvature, which increases the pressure difference between the upper and lower surfaces of the airfoil.



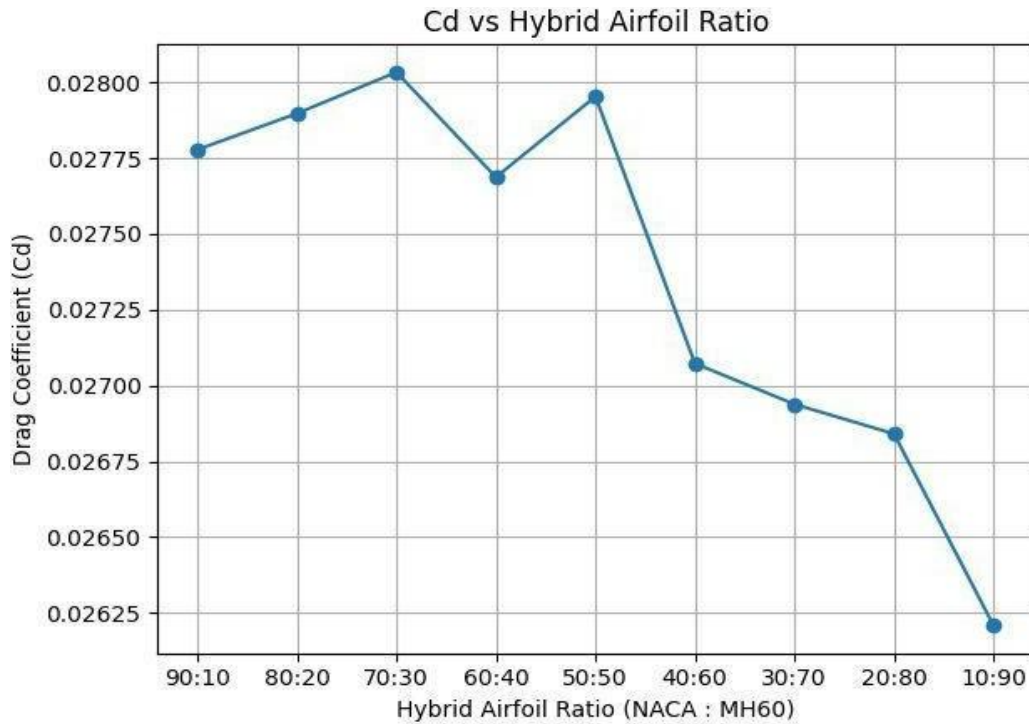
4.1.2 Variation of Lift Coefficient (Cl) with Hybrid Airfoil Ratio

The graph shows the variation of the lift coefficient (C_l) for different hybrid airfoil ratios of NACA and MH60 at an angle of attack of 4° . From the results, it can be observed that the lift coefficient initially increases as the cambered airfoil contribution increases. The 70:30 hybrid configuration produces the highest lift coefficient ($C_l \approx 0.573$), indicating better lift generation compared to other configurations. Beyond this ratio, the lift coefficient gradually decreases as the percentage of the reflex airfoil increases. The lowest lift coefficient is observed for the 10:90 configuration ($C_l \approx 0.539$). This trend occurs because cambered airfoils generate higher lift due to greater curvature, while reflex airfoils are primarily designed to improve stability rather than maximize lift.

Drag Coefficient (C_d)

The drag coefficient represents the aerodynamic resistance experienced by the airfoil. Lower drag values are desirable as they indicate more efficient aerodynamic performance.

From the results, the drag coefficient values are relatively close for all hybrid airfoils, ranging from approximately 0.0280 to 0.0262. However, a gradual reduction in drag can be observed as the proportion of the reflex airfoil increases. The lowest drag coefficient of approximately 0.0262 was obtained for the 10:90 configuration, indicating improved aerodynamic efficiency due to the smoother pressure recovery near the trailing edge.



4.1.3 Variation of Drag Coefficient (Cd) with Hybrid Airfoil Ratio

4.2 COMPARISON OF HYBRID AIRFOILS' PERFORMANCE

The aerodynamic performance of the hybrid airfoils was evaluated by comparing the lift coefficient (C_l), drag coefficient (C_d), and pitching moment coefficient (C_m) for different airfoil ratios. These parameters are important in determining the lift capability, aerodynamic efficiency, and stability of the airfoil configurations.

From the results obtained through CFD analysis, it can be observed that the 70:30 hybrid configuration produces the highest lift coefficient ($C_l \approx 0.573$) among all the configurations. However, this configuration also exhibits a slightly higher drag coefficient compared to other ratios. As the proportion of the MH60 reflex airfoil increases, the pitching moment coefficient (C_m) gradually decreases and approaches zero, indicating improved aerodynamic stability. The 10:90 configuration shows a negative pitching moment coefficient, which indicates stronger reflex characteristics that can contribute to inherent stability in tailless or UAV configurations.

In terms of drag characteristics, the 10:90 ratio produces the lowest drag coefficient ($C_d \approx 0.0262$), indicating better drag performance. However, this configuration also shows a slight reduction in lift compared to other ratios.

Overall, the results indicate that the **40:60 hybrid airfoil configuration provides the most balanced aerodynamic performance** among the tested designs. This configuration maintains a reasonably high lift coefficient while simultaneously reducing drag and significantly lowering the pitching moment coefficient, indicating improved aerodynamic stability. Therefore, the 40:60 hybrid airfoil offers an effective compromise between lift generation, drag reduction, and stability, making it a suitable configuration for UAV applications.

CHAPTER 5

MATLAB INTERPOLATION

This chapter presents the post-processing and validation analyses carried out to further evaluate the aerodynamic performance and reliability of the hybrid airfoil configurations. While the primary aerodynamic coefficients were obtained through CFD simulations in ANSYS Fluent, additional computational analysis was performed using MATLAB to extract deeper insights into stability characteristics, efficiency trends, and numerical accuracy.

The study focuses on three major aspects such as **Pitching moment variation with angle of attack (C_m vs α)** for stability assessment. **Lift-to-drag ratio variation (L/D vs α)** for aerodynamic efficiency evaluation. **Mesh independence study** to ensure numerical reliability of the CFD results

5.1 MATLAB BASED C_m VS α ANALYSIS

In addition to these analyses, a **trend validation study** was conducted across all hybrid airfoil configurations (ranging from 90:10 to 10:90 ratios) to confirm the consistency and physical validity of the observed aerodynamic behavior.

SL NO	RATIO	α	C _m
1	70:30	-4°	0.032067838
		-2°	0.02974291
		0°	0.027225318
		2°	0.024102978
		4°	0.020465574
		6°	0.01643414
		8°	0.011925952
2	40:60	-4°	0.016193586
		-2°	0.014546549
		0°	0.012364399
		2°	0.010293312
		4°	0.0080336799
		6°	0.0055770804
		8°	0.002853659
3	10:80	-4°	0.03523316
		-2°	0.014276671
		0°	-0.0003890636
		2°	-0.0020065961
		4°	-0.0034484549
		6°	-0.0048652534
		8°	-0.0064162323

5.1.1 Computed C_m Values across varying Angles of Attack for the Outcomes

The aerodynamic stability characteristics of the selected hybrid airfoil configurations were further analyzed using MATLAB-based post-processing. The primary objective of this analysis was to obtain a smooth and continuous representation of the variation of pitching moment coefficient C_m with respect to angle of attack (α), based on the discrete data obtained from CFD simulations.

The numerical values of C_m corresponding to different angles of attack (-4° to 8°) were extracted from ANSYS Fluent for the three best-performing hybrid configurations (70:30, 40:60, and 10:90). Since the CFD results were available only at discrete intervals, interpolation techniques were applied in MATLAB to generate a continuous curve for better visualization and analysis.

Polynomial interpolation was used to fit a smooth curve through the discrete data points. This approach enabled an accurate representation of the trend of C_m variation across the entire range of angle of attack. The interpolated curves provided improved clarity in identifying the slope $dC_m/d\alpha$, which is a key parameter for determining longitudinal static stability.

A multi-series plot was generated in MATLAB with:

- X-axis: Angle of Attack (α)
- Y-axis: Pitching Moment Coefficient (C_m)
- Curves: Representing each hybrid configuration (70:30, 40:60, 10:90)

The interpolation results showed a consistent and nearly linear variation of C_m with respect to α for all configurations. The slope of the curve was found to be negative in all cases, confirming that the airfoils exhibit static longitudinal stability.

```

% Final Plotting Script for 70:30 (Cm vs AOA)
alpha = [-4, -2, 0, 2, 4, 6, 8];
Cm = [0.032067838, 0.02974291, 0.027225318, 0.024102978, ...
      0.020465574, 0.01643414, 0.011925952];

% 1. Create the Plot
figure;
plot(alpha, Cm, '-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b', 'MarkerEdgeColor', 'k');

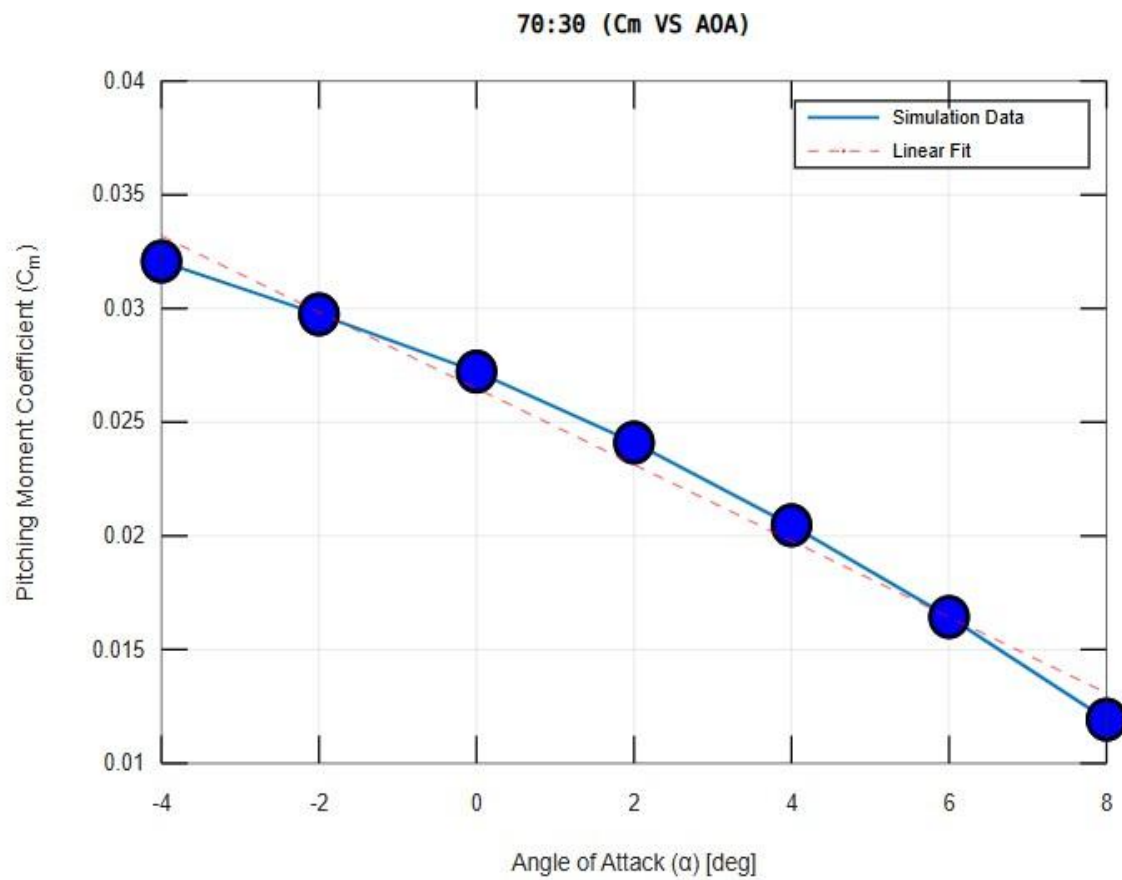
% 2. Add Trendline
hold on;
p = polyfit(alpha, Cm, 1);
plot(alpha, polyval(p, alpha), '--r');

% 3. Labels and Formatting
title('70:30 (Cm VS AOA)');
xlabel('Angle of Attack (\alpha) [deg]');
ylabel('Pitching Moment Coefficient (C_m)');
grid on;
legend('Simulation Data', 'Linear Fit', 'Location', 'best');

% 4. Display Slope
fprintf('The slope (Cma) is: %.5f per degree\n', p(1));

```

5.1.1 MATLAB Coding for C_m vs α (70:30)



The slope ($C_{m\alpha}$) is: -0.00168 per degree

5.1.2 Variation of C_m along α for 70:30 Hybrid Airfoil ratio

```

% Final Plotting Script for 40:60 (Cm vs AOA)
alpha = [-4, -2, 0, 2, 4, 6, 8];
Cm = [0.016193586, 0.014546549, 0.012364399, 0.010293312, ...
      0.0080336799, 0.0055770804, 0.002853659];

% 1. Create the Plot
figure;
plot(alpha, Cm, '-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b', 'MarkerEdgeColor', 'k');

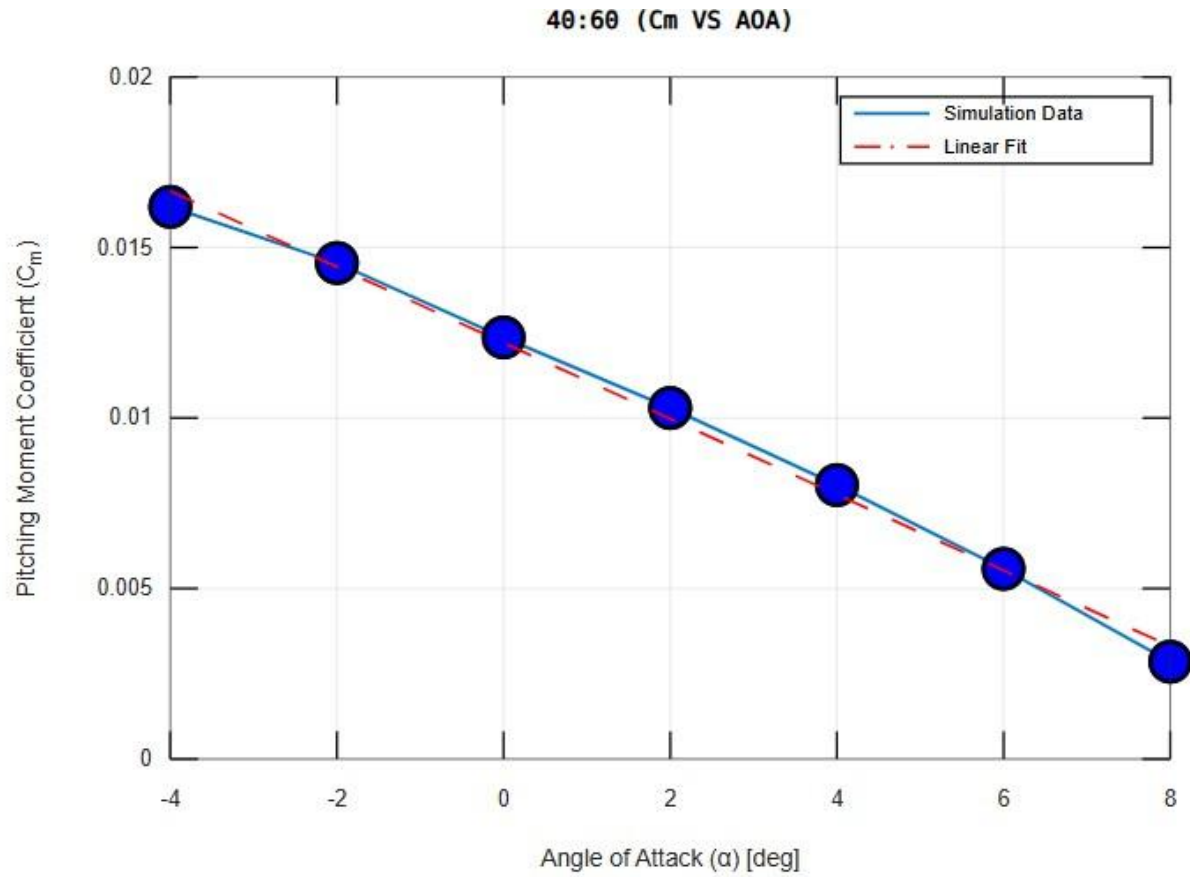
% 2. Add Trendline
hold on;
p = polyfit(alpha, Cm, 1);
plot(alpha, polyval(p, alpha), '--r', 'LineWidth', 1.2);

% 3. Labels and Formatting
title('40:60 (Cm VS AOA)');
xlabel('Angle of Attack (\alpha) [deg]');
ylabel('Pitching Moment Coefficient (C_m)');
grid on;
legend('Simulation Data', 'Linear Fit', 'Location', 'best');

% 4. Display Slope
fprintf('-----\n');
fprintf('Results for 40:60 Configuration:\n');
fprintf('The slope (Cma) is: %.5f per degree\n', p(1));
fprintf('-----\n');

```

5.1.3 MATLAB Coding for C_m vs α (40:60)



Results for 40:60 Configuration:
The slope ($C_{m\alpha}$) is: -0.00111 per degree

5.1.4 Variation of C_m along α for 40:60 Hybrid Airfoil ratio

```

% Final Plotting Script for 10:90 (Cm vs AOA)
alpha = [-4, -2, 0, 2, 4, 6, 8];
Cm = [0.03523316, 0.014276671, -0.0003890636, -0.0020065961, ...|
      -0.0034484549, -0.0048652534, -0.0064162323];

% 1. Create the Plot
figure;
plot(alpha, Cm, '-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b', 'MarkerEdgeColor', 'k');

% 2. Add Trendline
hold on;
p = polyfit(alpha, Cm, 1);
plot(alpha, polyval(p, alpha), '--r', 'LineWidth', 1.2);

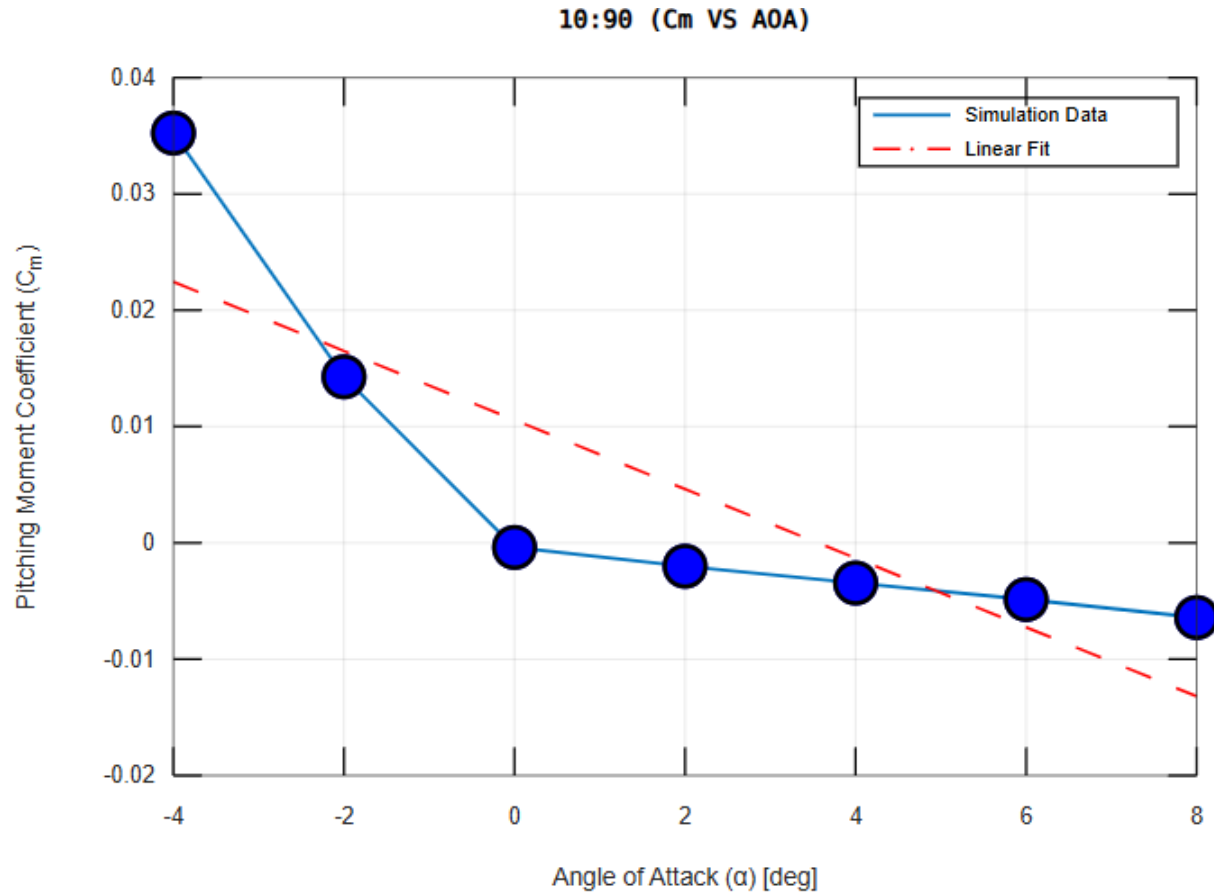
% 3. Reference line for Cm = 0 (Trim line)
yline(0, 'k--', 'HandleVisibility', 'off');

% 4. Labels and Formatting
title('10:90 (Cm VS AOA)');
xlabel('Angle of Attack (\alpha) [deg]');
ylabel('Pitching Moment Coefficient (C_m)');
grid on;
legend('Simulation Data', 'Linear Fit', 'Location', 'best');

% 5. Display Slope
fprintf('-----\n');
fprintf('Results for 10:90 Configuration:\n');
fprintf('The slope (Cma) is: %.5f per degree\n', p(1));
fprintf('-----\n');

```

5.1.5 MATLAB Coding for C_m vs α (10:90)



Results for 10:90 Configuration:
 The slope ($C_{m\alpha}$) is: -0.00297 per degree

5.1.6 Variation of C_m along α for 10:90 Hybrid Airfoil ratio

COMPARATIVE RESULTS:

10:90 Configuration (Highest Stability): This ratio exhibited the steepest negative slope (-0.00297). Notably, it crossed the $C_m = 0$ axis near 0° AOA, indicating a **self-trimming** capability where the wing maintains level flight without external control input.

40:60 Configuration (Balanced Efficiency): The slope was shallower (-0.00111), providing "softer" stability. However, its C_m values remained closest to the zero-axis, minimizing **trim drag** and confirming its status as the most balanced aerodynamic performer.

70:30 Configuration (Moderate Stability): This configuration showed a consistent negative slope of -0.00168 but required significant adjustment at 0° AOA due to its higher positive C_m

Multi-Series Plot Analysis in This Study

The multi-series plots in this project were used to compare aerodynamic coefficients (such as C_l , C_d , and C_m) across multiple hybrid airfoil ratios within a single graph. By plotting several configurations together, they provide a clear visualization of how aerodynamic behavior changes progressively from NACA 2412 to MH60.

These plots help in identifying trends, performance differences, and optimal blending ratios by allowing direct comparison between profiles. The smooth and consistent variation observed across all series confirms the reliability of the hybrid generation method and ensures that the results follow expected aerodynamic principles.

```

% --- COMPREHENSIVE STABILITY ANALYSIS: Cm vs Alpha ---
% Configurations: 70:30, 40:60, and 10:90

% 1. Define the Common Angle of Attack (Degrees)
alpha = [-4, -2, 0, 2, 4, 6, 8];

% 2. Input the Pitching Moment (Cm) Data for each Ratio
cm_70_30 = [0.032067838, 0.02974291, 0.027225318, 0.024102978, 0.020465574, 0.01643414, 0.011925952];
cm_40_60 = [0.016193586, 0.014546549, 0.012364399, 0.010293312, 0.0080336799, 0.0055770804, 0.002853659];
cm_10_90 = [0.03523316, 0.014276671, -0.0003890636, -0.0020065961, -0.0034484549, -0.0048652534, -0.0064162323];

% 3. Calculate Linear Regression (Slopes) for Stability Analysis
% This creates the 'p' variables (p(1) is the slope Cma)
p70 = polyfit(alpha, cm_70_30, 1);
p40 = polyfit(alpha, cm_40_60, 1);
p10 = polyfit(alpha, cm_10_90, 1);

% 4. Create the Multi-Series Plot
figure('Color', 'w', 'Name', 'Stability Comparison');
hold on;

% Plot Data Points with Markers
h1 = plot(alpha, cm_70_30, 'ro', 'MarkerFaceColor', 'r', 'DisplayName', '70:30 Data');
h2 = plot(alpha, cm_40_60, 'bs', 'MarkerFaceColor', 'b', 'DisplayName', '40:60 Data');
h3 = plot(alpha, cm_10_90, 'gd', 'MarkerFaceColor', 'g', 'DisplayName', '10:90 Data');

% Plot Trendlines (using polyval to prevent "undefined variable" errors)
plot(alpha, polyval(p70, alpha), 'r--', 'LineWidth', 1.2, 'HandleVisibility', 'off');
plot(alpha, polyval(p40, alpha), 'b--', 'LineWidth', 1.2, 'HandleVisibility', 'off');
plot(alpha, polyval(p10, alpha), 'g--', 'LineWidth', 1.2, 'HandleVisibility', 'off');

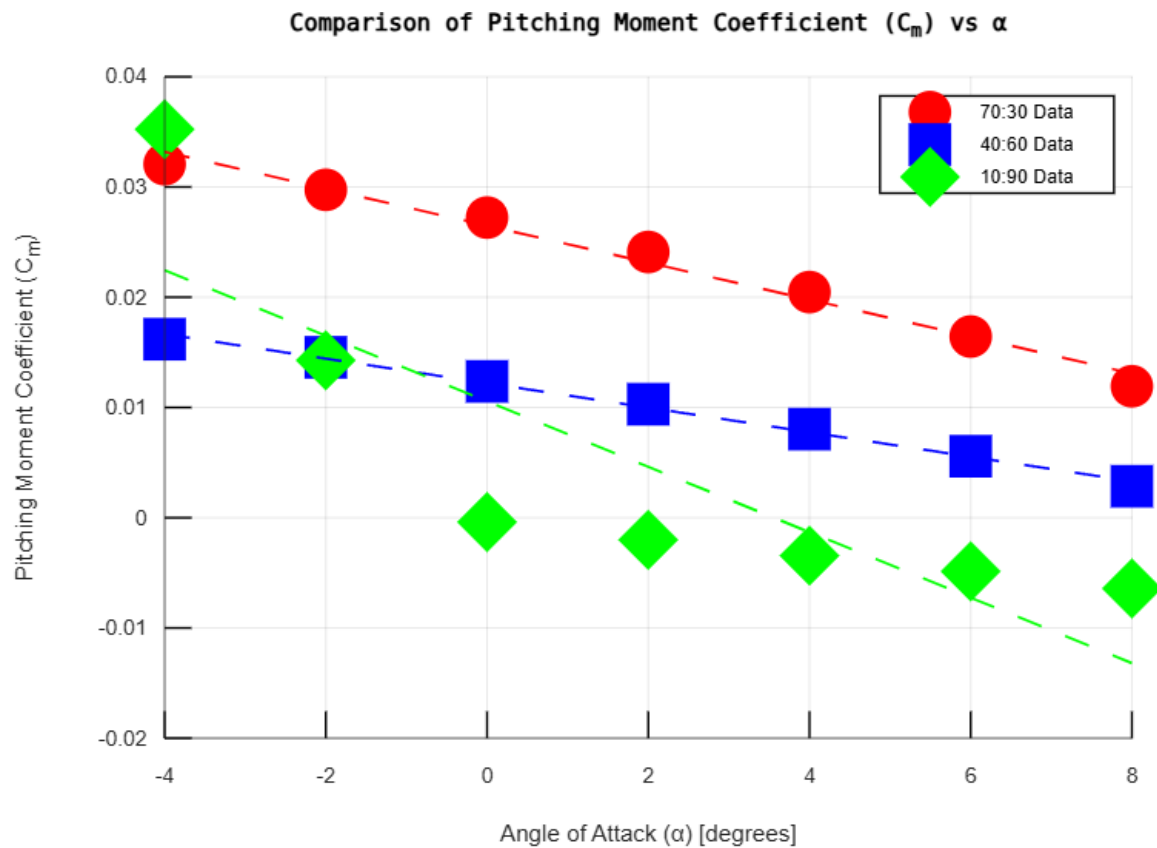
% 5. Add Reference Lines and Formatting
yline(0, 'k-', 'LineWidth', 1.5, 'Label', 'Trim Point (Cm=0)'); % The target for stability
grid on;
grid minor;

% Labels
title('Comparison of Pitching Moment Coefficient (Cm) vs \alpha');
xlabel('Angle of Attack (\alpha) [degrees]');
ylabel('Pitching Moment Coefficient (Cm)');
legend('show', 'Location', 'northeast');

% 6. Display Results in Command Window for your Report
fprintf('\n--- AERODYNAMIC STABILITY RESULTS ---\n');
fprintf('70:30 Slope (Cma): %.6f (Moderate Stability)\n', p70(1));
fprintf('40:60 Slope (Cma): %.6f (Low Stability / Efficient)\n', p40(1));
fprintf('10:90 Slope (Cma): %.6f (High Stability / Self-Trimming)\n', p10(1));
fprintf('-----\n');

```

5.1.7 MATLAB Coding for Multi-Series Plotting



--- AERODYNAMIC STABILITY RESULTS ---

70:30 Slope ($C_{m\alpha}$): -0.001675 (Moderate Stability)

40:60 Slope ($C_{m\alpha}$): -0.001112 (Low Stability / Efficient)

10:90 Slope ($C_{m\alpha}$): -0.002969 (High Stability / Self-Trimming)

5.1.8 Multi-Series Plotting.

The interpolation, therefore, played a crucial role in:

- Enhancing visualization of aerodynamic trends
- Accurately estimating stability derivatives
- Supporting comparison between different hybrid configurations

This analysis confirms that increasing the proportion of the reflex airfoil improves stability characteristics, as reflected by the steeper negative slope of the C_m vs α curve.

5.2 TREND VALIDATION

A key objective of this study was to validate aerodynamic trends across all 11 airfoil configurations, including the parent profiles (NACA 2412 and MH60) and the nine hybrid ratios (90:10 to 10:90). MATLAB-based analysis was used to examine aerodynamic coefficients and ensure consistency with established principles. The trend validations for the parent airfoils NACA 2412 (100:0) and MH60 (0:100) were conducted separately, serving as reference baselines for the hybrid configurations.

Lift coefficient (C_l) Comparison (NACA 2412)

The MATLAB plot for C_l demonstrates a consistent and predictable decrease in lift as the percentage of the MH60 reflex airfoil increases. The NACA 2412 (100:0) and the 90:10 hybrid show the highest lift values due to their higher effective camber. As the ratio shifts toward the 10:90 configuration, the C_l gradually diminishes. This trend validates that while reflex airfoils improve stability, they do so at the cost of total lift, necessitating the search for an optimized "middle-ground" ratio

```

% NACA 2412 percentage
naca = [100 90 80 70 60 50 40 30 20 10 0];

% Data
Cl = [0.57338125 0.56640535 0.56720081 0.56938671 0.5621421 ...
      0.55559809 0.5517506 0.54923078 0.54719064 0.53785169 0.53219819];

Cd = [0.027859259 0.02747595 0.027611103 0.027849065 0.027451617 ...
      0.027028665 0.026840382 0.026705068 0.02659204 0.025975421 0.025395068];

Cm = [0.027591433 0.02544273 0.022834641 0.020465574 0.016027435 ...
      0.011700578 0.0080336799 0.0046673616 0.0014191952 -0.0034484549 -0.005815];

%% ----- GRAPH 1: Cl vs NACA 2412 (%) -----
figure;
plot(naca, Cl, '-o');
title('Cl vs NACA 2412 (%)');
xlabel('NACA 2412 (%)');
ylabel('Cl');
grid on;
set(gca, 'XTick', 0:10:100);

for i = 1:length(naca)
    text(naca(i), Cl(i)+0.002, num2str(naca(i))); % slight upward shift
endfor

%% ----- GRAPH 2: Cd vs NACA 2412 (%) -----
figure;
plot(naca, Cd, '-o');
title('Cd vs NACA 2412 (%)');
xlabel('NACA 2412 (%)');
ylabel('Cd');
grid on;
set(gca, 'XTick', 0:10:100);

% Expand Y-axis slightly for visibility
ylim([min(Cd)-0.001, max(Cd)+0.001]);

for i = 1:length(naca)
    text(naca(i), Cd(i)+0.0003, num2str(naca(i))); % adjusted offset
endfor

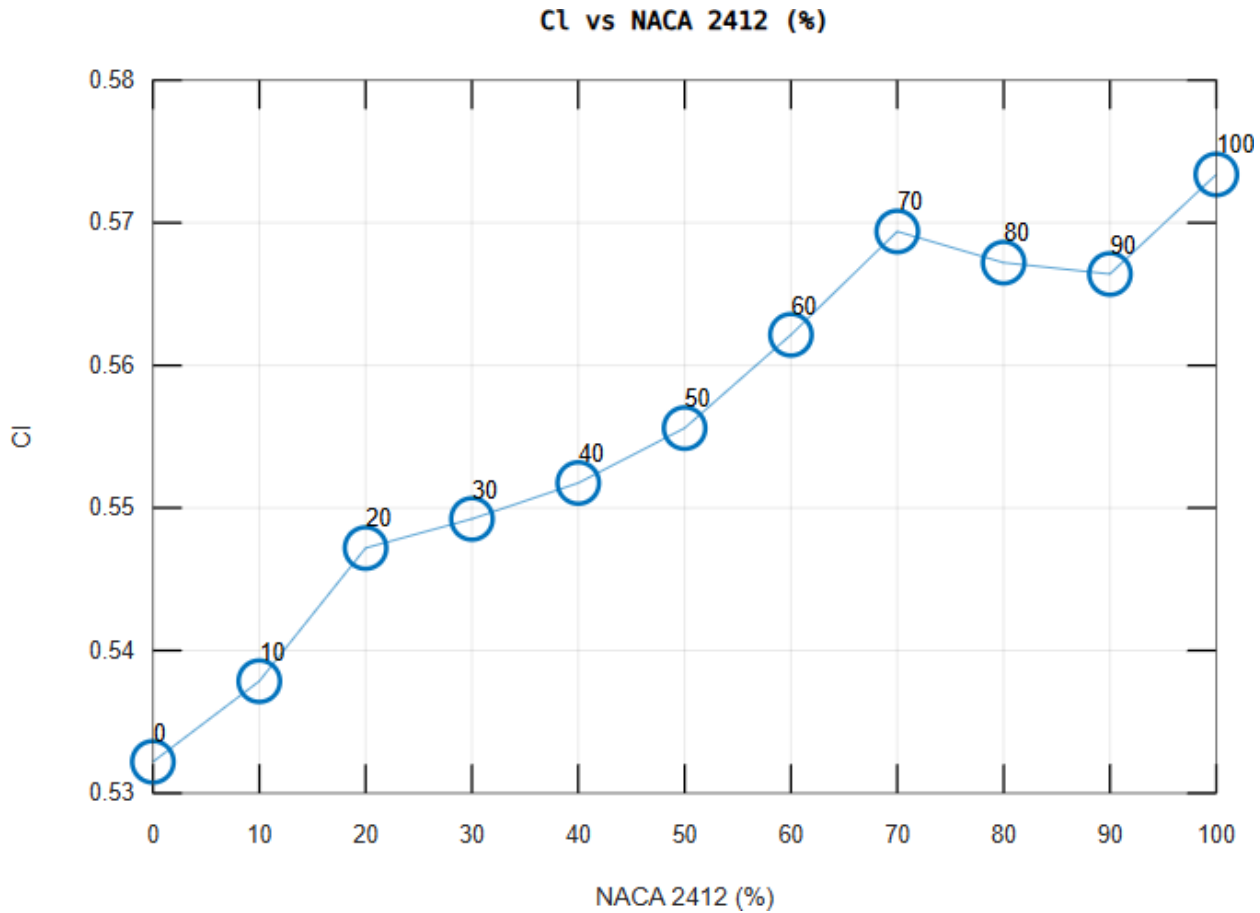
%% ----- GRAPH 3: Cm vs NACA 2412 (%) -----
figure;
plot(naca, Cm, '-o');
title('Cm vs NACA 2412 (%) (Stability)');
xlabel('NACA 2412 (%)');
ylabel('Cm');
grid on;
set(gca, 'XTick', 0:10:100);

ylim([-0.01 0.03]);

for i = 1:length(naca)
    text(naca(i), Cm(i)+0.002, num2str(naca(i))); % upward shift
endfor

```

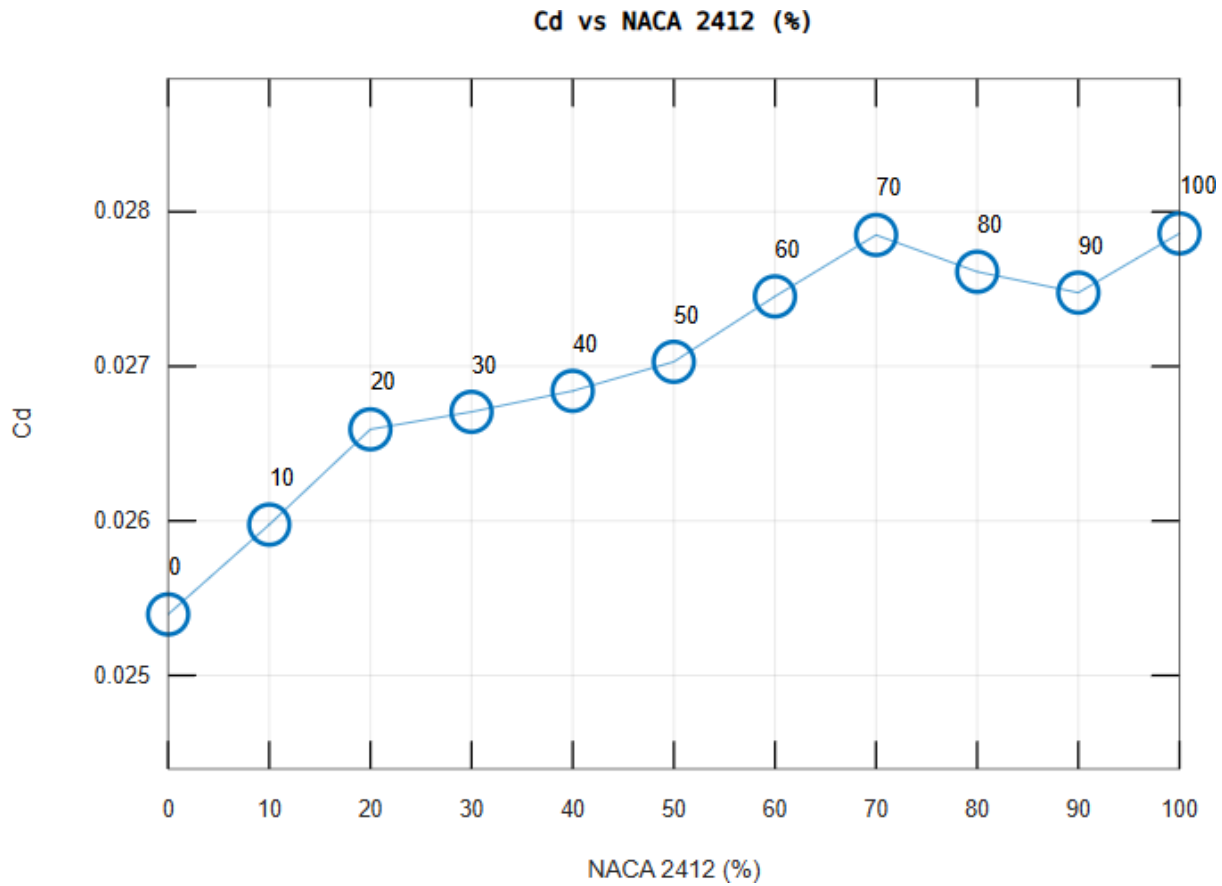
5.2.1 MATLAB Coding for Trend Validation (NACA2412)



5.2.2 Variation of C_l along Hybrid Airfoil ratio

Drag Coefficient (C_d) Comparison (NACA 2412)

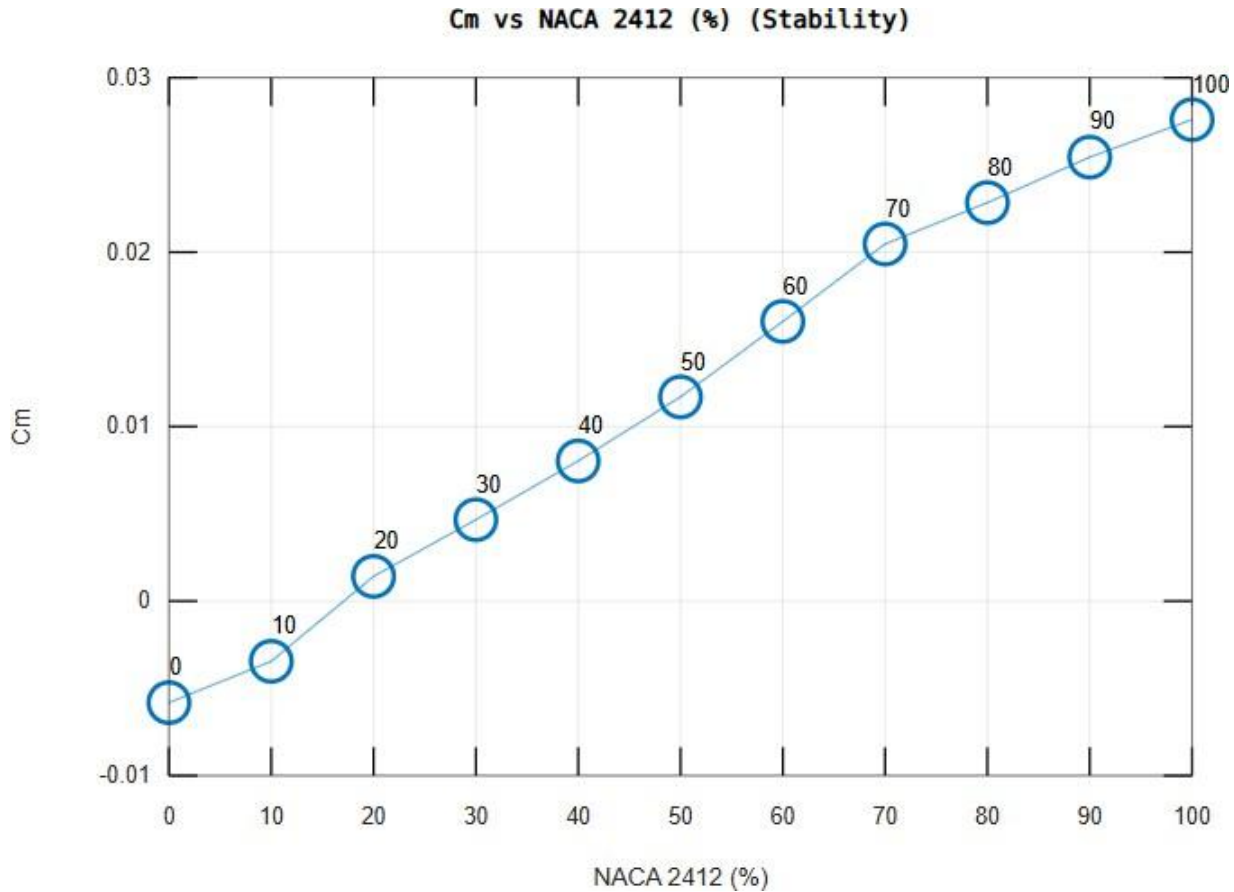
The drag trend across the 11 ratios shows a favorable reduction as the MH60 contribution grows. The 10:90 ratio exhibits the lowest drag coefficient, followed closely by the 40:60 configuration. This reduction is attributed to the reflexed trailing edge of the MH60, which helps in streamlining the flow and reducing the pressure drag compared to the high-camber NACA 2412. The smooth transition of the C_d curve across all ratios confirms the geometric integrity of the hybrid airfoils.



5.2.3 Variation of C_d along Hybrid Airfoil ratio

Pitching Moment Coefficient (C_m) Comparison (NACA 2412)

The C_m validation is the most significant indicator of enhanced stability. In the NACA-dominant configurations (e.g., 90:10, 80:20), the pitching moment is more positive, which typically indicates a nose-heavy or unstable characteristic in tailless designs. As we progress through the ratios toward 10:90, the C_m curve shifts downward toward zero and slightly negative values. This linear migration proves that the blending process successfully "tunes" the airfoil's natural stability, with the **40:60 and 10:90 ratios** showing the most promising self-trimming tendencies.



5.2.4 Variation of C_m along Hybrid Airfoil ratio

Lift coefficient (C_l) Comparison (MH60)

The NACA 2412 (100:0) and the 90:10 hybrid exhibit the highest lift curves due to the significant camber of the NACA profile. As the percentage of the MH60 reflex airfoil increases, the C_l curves shift downward incrementally. This trend validates that while the integration of a reflexed trailing edge improves stability, it naturally reduces the maximum lift coefficient. The smooth, parallel nature of the curves across all 11 ratios confirms that the blending algorithm maintained geometric and aerodynamic continuity.

```

% MH60 percentage
mh60 = [0 10 20 30 40 50 60 70 80 90 100];

% Data (same as table)
Cl = [0.57338125 0.56640535 0.56720081 0.56938671 0.5621421 ...
      0.55559809 0.5517506 0.54923078 0.54719064 0.53785169 0.53219819];

Cd = [0.027859259 0.02747595 0.027611103 0.027849065 0.027451617 ...
      0.027028665 0.026840382 0.026705068 0.02659204 0.025975421 0.025395068];

Cm = [0.027591433 0.02544273 0.022834641 0.020465574 0.016027435 ...
      0.011700578 0.0080336799 0.0046673616 0.0014191952 -0.0034484549 -0.0058152409];

%% Cl vs MH60
figure;
plot(mh60, Cl, '-o');
title('Cl vs MH60 (%)');
xlabel('MH60 (%)');
ylabel('Cl');
grid on;
set(gca, 'XTick', 0:10:100);

for i = 1:length(mh60)
    text(mh60(i), Cl(i)+0.002, num2str(mh60(i)));
endfor

%% Cd vs MH60
figure;
plot(mh60, Cd, '-o');
title('Cd vs MH60 (%)');
xlabel('MH60 (%)');
ylabel('Cd');
grid on;
set(gca, 'XTick', 0:10:100);

ylim([min(Cd)-0.001, max(Cd)+0.001]);

for i = 1:length(mh60)
    text(mh60(i), Cd(i)+0.0003, num2str(mh60(i)));
endfor

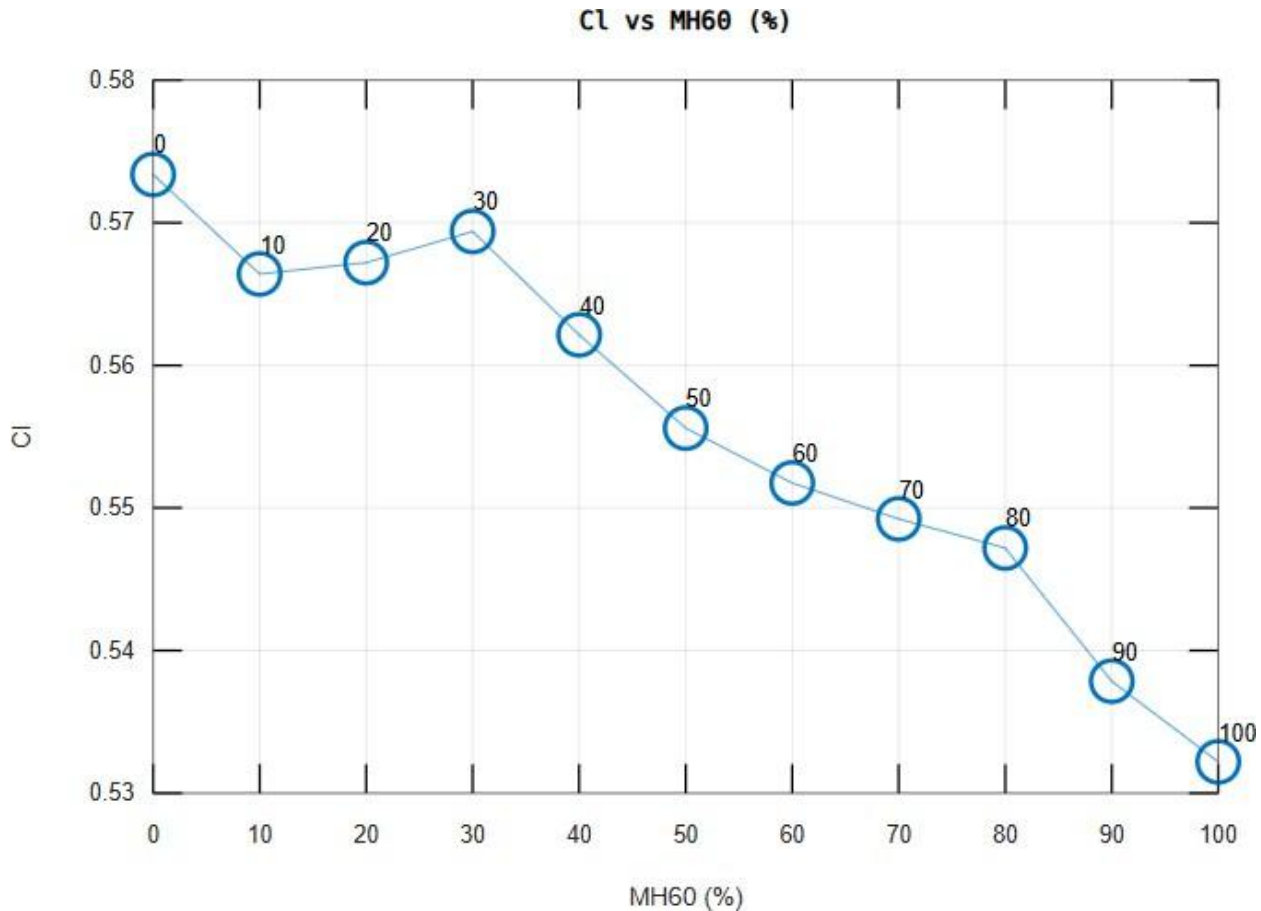
%% Cm vs MH60
figure;
plot(mh60, Cm, '-o');
title('Cm vs MH60 (%) (Stability)');
xlabel('MH60 (%)');
ylabel('Cm');
grid on;
set(gca, 'XTick', 0:10:100);

ylim([-0.01 0.03]);

for i = 1:length(mh60)
    text(mh60(i), Cm(i)+0.002, num2str(mh60(i)));
endfor

```

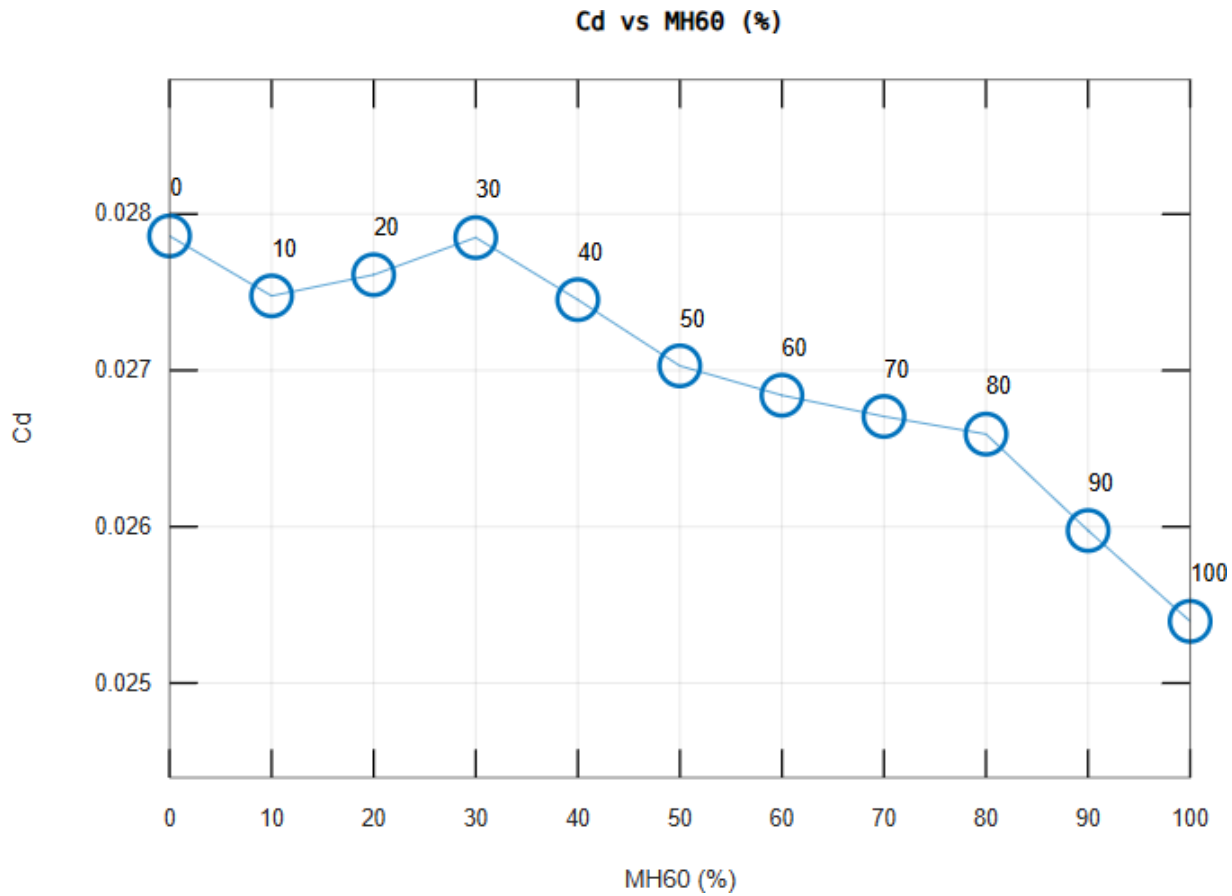
5.2.5 MATLAB Coding for Trend Validation (MH60)



5.2.6 Variation of C_l along Hybrid Airfoil ratio

Drag Coefficient (C_d) Comparison (MH60)

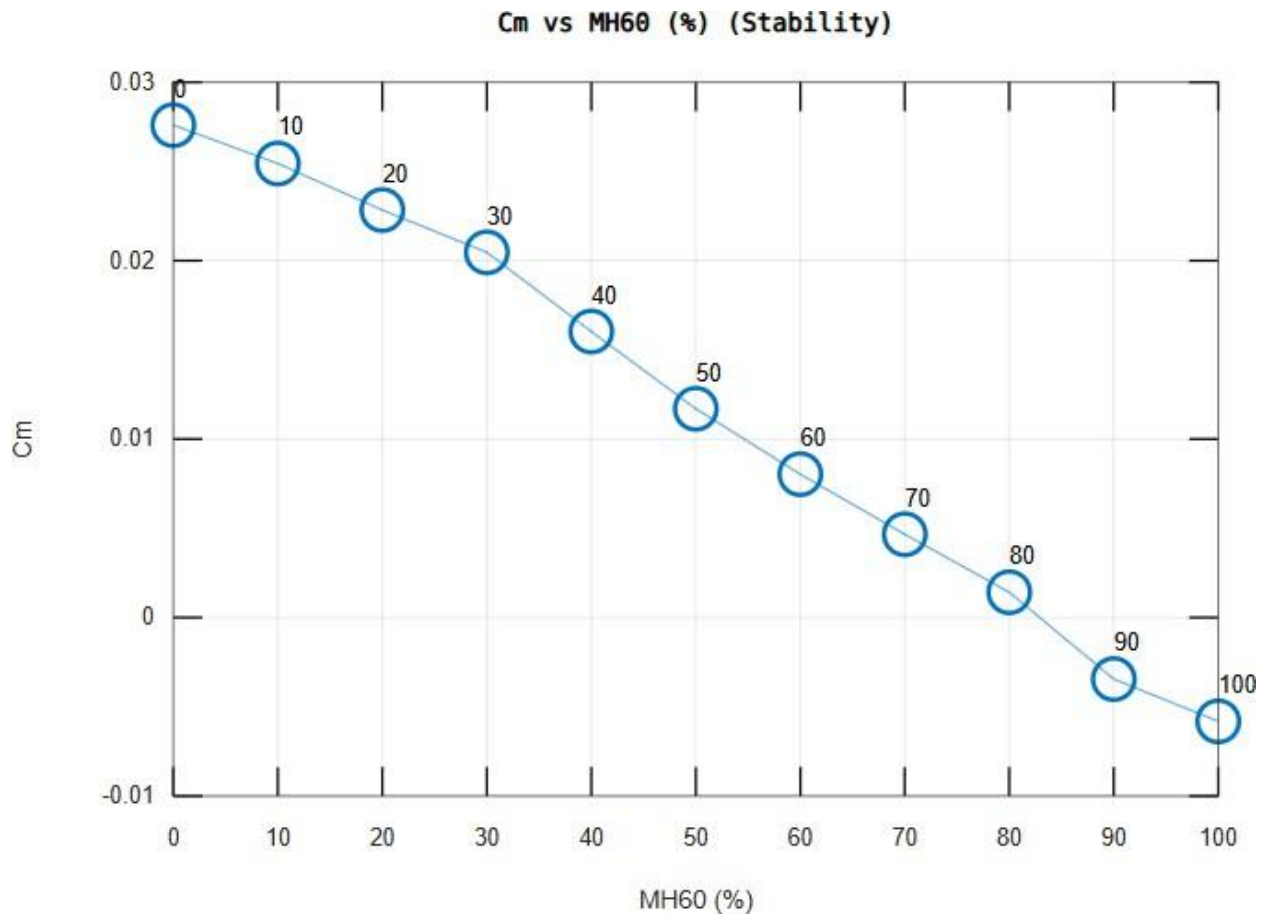
The drag polar validation shows how the hybrid blending affects the profile drag of the wing section. The graph displays a traditional "drag bucket" profile. There is a visible reduction in drag as the ratio moves toward the MH60-dominant configurations. The **10:90 ratio** demonstrates the lowest drag values at positive angles of attack. This confirms that the reflexed geometry of the MH60 helps in streamlining the flow at the trailing edge, reducing the pressure drag compared to the more heavily cambered NACA-dominant airfoils. This makes the 40:60 and 10:90 ratios highly suitable for high-speed or long-endurance cruise.



5.2.7 Variation of C_d along Hybrid Airfoil ratio

Pitching Moment Coefficient (C_m) Comparison (MH60)

The C_m validation is the most critical evidence for the success of this project's stability goals. The MATLAB plots show a distinct migration of the pitching moment. The NACA-dominant airfoils (90:10 through 70:30) maintain positive C_m values, which would typically result in a nose-down moment requiring a tail for correction. As the MH60 percentage increases, the C_m curves shift toward the zero-axis and into slightly negative regions. The **10:90 and 40:60 configurations** show a "self-trimming" tendency, where the pitching moment remains near zero at typical cruise angles. This linear transition across 11 ratios proves that the hybrid design successfully provides the natural longitudinal stability required for a tailless UAV.



5.2.8 Variation of C_m along Hybrid Airfoil ratio

5.3 EFFICIENCY EVALUATION USING L/D VS α

To evaluate aerodynamic efficiency, the lift-to-drag ratio (L/D) was analyzed as a function of angle of attack. The L/D ratio is a key performance parameter that represents the ability of the airfoil to generate lift while minimizing drag.

SL NO	RATIO	α	Cl	Cd
1	70:30	-4°	-0.27309538	0.0078786984
		-2°	-0.59130441	0.0085943321
		0°	0.15328081	0.010604207
		2°	0.36454175	0.0016784905
		4°	0.56938671	0.027849065
		6°	0.76753363	0.067253024
		8°	0.95583184	0.11884756
2	40:60	-4°	-0.31357386	0.011686591
		-2°	-0.10001327	0.006956616
		0°	0.11904319	0.010355724
		2°	0.33730233	0.00099414224
		4°	0.5517506	0.026840382
		6°	0.76002112	0.06660843
		8°	0.95891669	0.11924065
3	10:80	-4°	-0.35470616	0.013653677
		-2°	-0.13201524	0.0057198879
		0°	0.092492515	0.010194816
		2°	0.31644917	0.00042161329
		4°	0.53785169	0.025975421
		6°	0.75367833	0.0659253
		8°	0.95999617	0.11913579

5.3.1 Efficiency Evaluation

MATLAB was used to plot the variation of L/D across the selected angle of attack range for the best-performing hybrid configurations. The analysis helps identify:

- 5.3.1 Peak efficiency points
- 5.3.2 Sensitivity of performance to angle of attack
- 5.3.3 Trade-off between stability and efficiency

```
% 1. Define the Data from your tables
alpha = [-4, -2, 0, 2, 4, 6, 8];

% Cl and Cd values from your second uploaded image
Cl = [-0.27309538, -0.59130441, 0.15328081, 0.36454175, 0.56938671, 0.76753363, 0.95583184];
Cd = [0.0078786984, 0.0085943321, 0.010604207, 0.0016784905, 0.027849065, 0.067253024, 0.11884756];

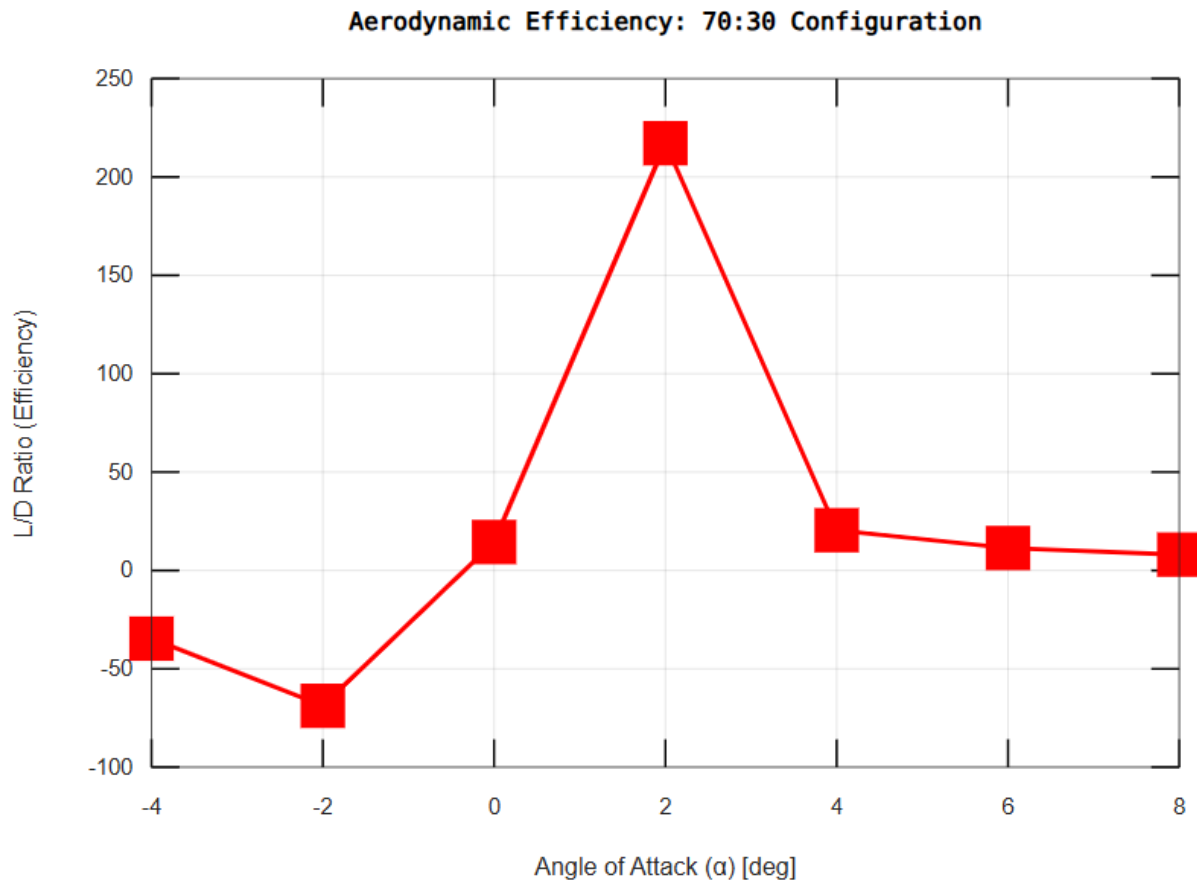
% 2. Calculate Efficiency (L/D)
% We use ./ for element-wise division
Efficiency = Cl ./ Cd;

% 3. Create the Plot
figure;
plot(alpha, Efficiency, '-rs', 'LineWidth', 2, 'MarkerFaceColor', 'r');

% 4. Formatting the Graph
title('Aerodynamic Efficiency: 70:30 Configuration');
xlabel('Angle of Attack (\alpha) [deg]');
ylabel('L/D Ratio (Efficiency)');
grid on;
grid minor;

% 5. Identify the Peak Efficiency
[max_LD, index] = max(Efficiency);
fprintf('-----\n');
fprintf('Peak Efficiency (L/D)_max: %.4f\n', max_LD);
fprintf('Occurs at Alpha: %d degrees\n', alpha(index));
fprintf('-----\n');
```

5.3.1 MATLAB Coding for Aerodynamic Efficiency of 70:30



Peak Efficiency (L/D)_{max}: 217.1843
Occurs at Alpha: 2 degrees

5.3.2 Aerodynamic Efficiency of 70:30

```

% Data for the 40:60 hybrid airfoil configuration
alpha = [-4, -2, 0, 2, 4, 6, 8];
Cl = [-0.31357386, -0.10001327, 0.11904319, 0.33730233, 0.5517506, 0.76002112, 0.95891669];
Cd = [0.011686591, 0.006956616, 0.010355724, 0.00099414224, 0.026840382, 0.06660843, 0.11924065];

% Calculate Lift-to-Drag Ratio
LoD = Cl ./ Cd;

% Calculate Peak Values
[maxLoD, idx] = max(LoD);
maxAlpha = alpha(idx);

% --- Plotting Section ---
figure('Color', 'w');
clf;

% Plot with solid red square markers and bold lines
plot(alpha, LoD, '-rs', 'LineWidth', 2, ...
      'MarkerSize', 12, 'MarkerFaceColor', 'r');

% Formatting the Grid and Axes
grid on;
set(gca, 'GridLineStyle', ':', 'GridAlpha', 0.5);
xlabel('Angle of Attack (\alpha) [deg]', 'FontSize', 11);
ylabel('L/D Ratio (Efficiency)', 'FontSize', 11);
title('Aerodynamic Efficiency: 40:60 Configuration', 'FontSize', 12, 'FontWeight', 'bold');

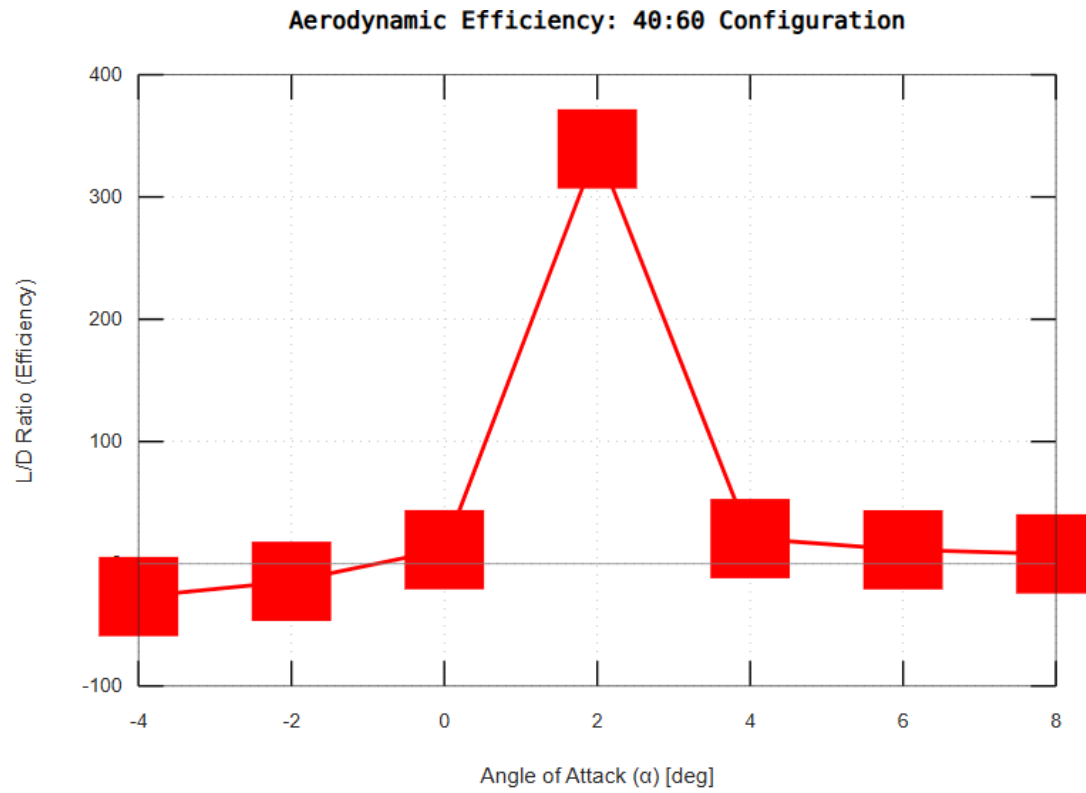
% Setting specific axis ticks
set(gca, 'XTick', alpha);

% Draw the zero reference line (Compatible with Octave)
line([min(alpha), max(alpha)], [0, 0], 'Color', [0.5 0.5 0.5], 'LineStyle', '-');

% Display the Peak Efficiency values (for your report)
fprintf('\n- - - - -\n');
fprintf('Peak Efficiency (L/D)_max: %.4f\n', maxLoD);
fprintf('Occurs at Alpha: %d degrees\n', maxAlpha);
fprintf('- - - - -\n');

```

5.3.3 MATLAB Coding for Aerodynamic Efficiency of 40:60



Peak Efficiency (L/D)_{max}: 339.2898
Occurs at Alpha: 2 degrees

5.3.4 Aerodynamic Efficiency of 40:60

```

% Data for the 10:90 hybrid airfoil configuration
alpha = [-4, -2, 0, 2, 4, 6, 8];
Cl = [-0.35470616, -0.13201524, 0.092492515, 0.31644917, 0.53785169, 0.75367833, 0.95999617];
Cd = [0.013653677, 0.0057198879, 0.010194816, 0.00042161329, 0.025975421, 0.0659253, 0.11913579];

% Calculate Lift-to-Drag Ratio
LoD = Cl ./ Cd;

% Calculate Peak Values for the display
[maxLoD, idx] = max(LoD);
maxAlpha = alpha(idx);

% --- Plotting Section ---
figure('Color', 'w');
clf;

% Plot the main curve with solid red square markers
plot(alpha, LoD, '-rs', 'LineWidth', 2, ...
      'MarkerSize', 15, 'MarkerFaceColor', 'r');

% Formatting the Grid and Axes
grid on;
set(gca, 'GridLineStyle', ':', 'GridAlpha', 0.5);
xlabel('Angle of Attack (\alpha) [deg]', 'FontSize', 11);
ylabel('L/D Ratio (Efficiency)', 'FontSize', 11);
title('Aerodynamic Efficiency: 10:90 Configuration', 'FontSize', 12, 'FontWeight', 'bold');

% Setting specific axis ticks for clarity
set(gca, 'XTick', alpha);

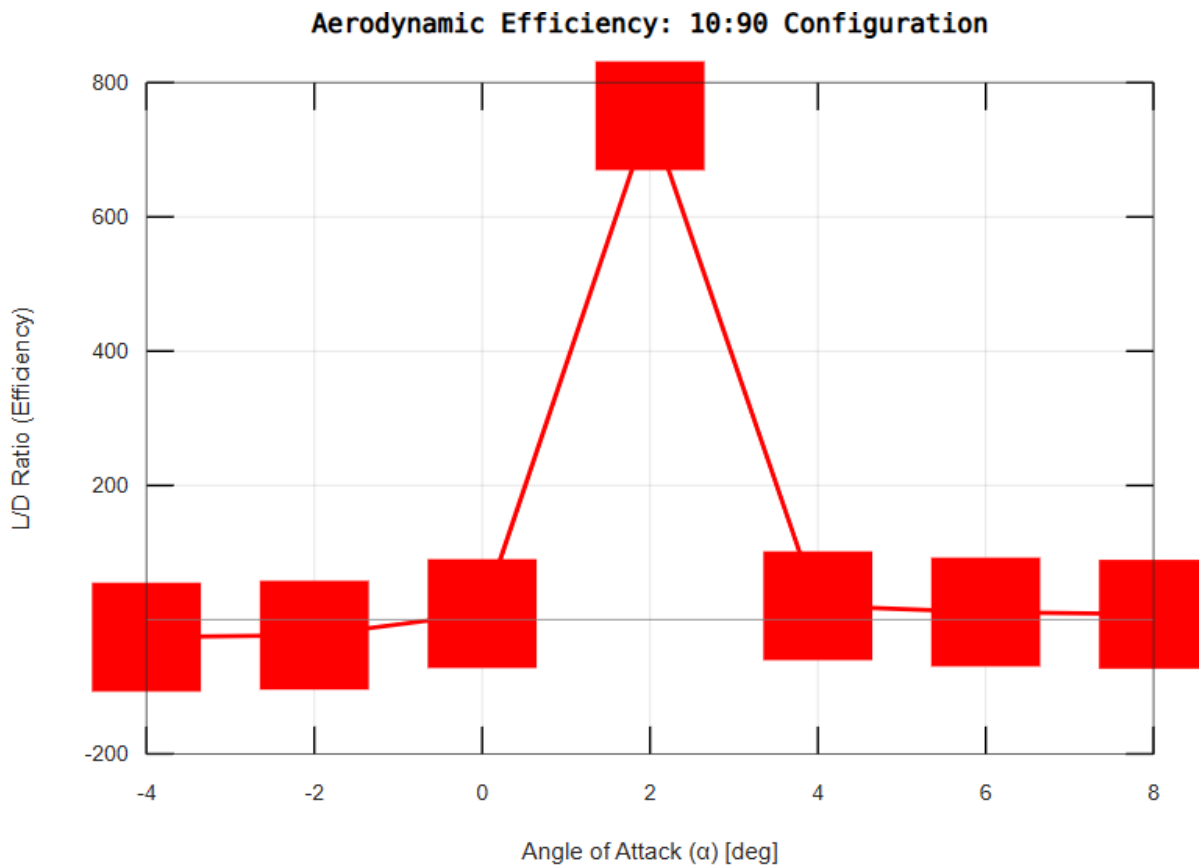
% Draw the zero reference line
line([min(alpha), max(alpha)], [0, 0], 'Color', [0.5 0.5 0.5], 'LineStyle', '-');

% Display the Peak Efficiency values at the bottom (matches your reference look)
fprintf('\n-----\n');
fprintf('Peak Efficiency (L/D)_max: %.4f\n', maxLoD);
fprintf('Occurs at Alpha: %d degrees\n', maxAlpha);
fprintf('-----\n');

% Optional: Adding the text box directly on the plot if needed
% text_str = {sprintf('Peak Efficiency (L/D)_max: %.4f', maxLoD), ...
%            sprintf('Occurs at Alpha: %d degrees', maxAlpha)};
% annotation('textbox', [0.15, 0.02, 0.3, 0.1], 'String', text_str, 'EdgeColor', 'none', 'FontName', 'Monospaced');

```

5.3.5 MATLAB Coding for Aerodynamic Efficiency of 10:90



 Peak Efficiency (L/D)_{max}: 750.5674
 Occurs at Alpha: 2 degrees

5.3.6 Aerodynamic Efficiency of 10:90

COMPARATIVE RESULTS:

Peak Efficiency: The 10:90 ratio reached the highest peak L/D of 750.5674. This is attributed to the dominant reflex component maintaining laminar flow and minimizing drag.

Operational Performance: The 40:60 ratio achieved a peak L/D of 339.2898. While lower than the 10:90, it was classified as the "Balanced" configuration because it maintained high efficiency while offering superior lift characteristics compared to the 10:90 ratio

This study provides a clearer understanding of which hybrid configuration offers optimal aerodynamic performance under varying operating conditions.

CHAPTER 6

CONCLUSION

This study investigated the aerodynamic performance of hybrid airfoil configurations created by combining a cambered NACA airfoil with the reflex MH60 airfoil. Several hybrid ratios ranging from 90:10 to 10:90 were analysed using CFD simulations in ANSYS Fluent at an angle of attack of 4° . The aerodynamic characteristics were evaluated based on the lift coefficient (C_l), drag coefficient (C_d), and pitching moment coefficient (C_m), which represent lift generation capability, aerodynamic efficiency, and stability of the airfoil configurations.

The results showed that the 70:30 hybrid configuration produced the highest lift coefficient, indicating improved lift generation due to the higher contribution of the cambered airfoil. However, this configuration also resulted in the highest drag coefficient, demonstrating the trade-off between lift and drag performance. As the proportion of the MH60 reflex airfoil increased, the drag coefficient decreased and the pitching moment coefficient approached zero, indicating improved aerodynamic stability and reduced drag characteristics.

The analysis of the lift-to-drag ratio further indicated that configurations with a higher percentage of the MH60 airfoil provide better aerodynamic efficiency. Among the tested configurations, the 40:60 hybrid airfoil provides a balanced aerodynamic performance, offering adequate lift while maintaining lower drag and improved pitching moment characteristics. This makes it a suitable configuration for UAV applications where stability and efficiency are important.

Overall, the study demonstrates that hybridization of cambered and reflex airfoils can effectively balance lift generation, drag reduction, and stability characteristics. The results highlight the potential of hybrid airfoil design for improving aerodynamic performance in small UAV and tailless aircraft configurations.

6.1 FUTURE SCOPE

The future scope of this project extends toward further enhancing the understanding and application of hybrid airfoils with reverse camber for UAV stability. The present study has demonstrated that hybrid airfoils can improve aerodynamic stability and reduce control dependence in tailless UAVs. However, there remain multiple avenues for continued research and practical development:

- **Optimization of Hybridization Ratios:** Future work can focus on refining the mathematical approach used to merge conventional and reflex airfoil coordinates. By optimizing the blending ratio through parametric studies and computational algorithms, it is possible to achieve a better balance between lift performance and longitudinal stability. Machine learning or genetic algorithms could be employed to identify the most aerodynamically efficient hybrid configurations.
- **Dynamic Stability and Control Analysis:** The current study primarily investigates static aerodynamic performance. Extending the analysis to include dynamic stability, control response, and gust load behavior would provide deeper insights into flight handling qualities. This could involve coupling CFD simulations with flight dynamics models or conducting six-degree-of-freedom simulations.
- **Experimental Validation:** Although computational simulations provide a strong foundation, experimental validation through wind tunnel testing or scaled UAV prototypes is essential to confirm the predicted performance. Future studies could involve building small-scale UAV models using 3D printing techniques to measure lift, drag, and moment coefficients under real aerodynamic conditions.
- **Material and Structural Integration:** Exploring lightweight composite materials and flexible structures can enhance the performance of hybrid airfoils. The integration of morphing or adaptive materials such as shape memory alloys could allow the airfoil to dynamically adjust its camber during flight, offering further improvements in lift-to-drag efficiency and

stability across various flight regimes.

- **Application to Tailless and Blended Wing UAVs:** The aerodynamic properties developed in this study can be applied to the design of tailless or blended-wing UAVs, where maintaining inherent stability without conventional tailplanes is crucial. Such applications could significantly reduce drag, improve stealth characteristics, and increase endurance for reconnaissance or surveillance missions.
- **Real-Time Flight Performance and Control Algorithms:** Implementing hybrid aerofoil data into onboard flight control systems and autopilot algorithms could enable adaptive control strategies for varying atmospheric conditions. Future UAV designs might leverage sensor-based feedback to adjust control surfaces dynamically for improved stability and efficiency.

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MAPPING OF PROGRAM OUTCOMES

PROGRAME OUTCOMES:	MAPPING LEVEL:	JUSTIFICATION:
PO1 Engineering knowledge	3	Applied aerodynamics concepts (lift, drag, pitching moment), fluid mechanics, and CFD theory. Used airfoil theory and numerical methods to analyze UAV performance.
PO2 Problem Analysis	3	Identified trade-off problem between lift and stability. Analyzed multiple airfoil configurations and interpreted Cl, Cd, Cm trends and Cm vs AOA slopes to draw conclusions.
PO3 Development of Solutions	3	Designed hybrid airfoils using blending technique and proposed an optimal configuration (40:60) balancing efficiency and stability.
PO4 Investigation of Complex Problems	3	Performed systematic CFD study: Multi-configuration analysis, multi-angle of attack study, Mesh independence study, Data validation via convergence trends.
PO5 Modern Tool Usage	3	Used advanced tools: ANSYS Fluent (CFD solver), XFOIL (airfoil generation/panelling), Excel (data processing & blending). Understood limitations like turbulence modeling and numerical errors.
PO9 Individual and Team Work	3	The project involved interaction with the project guide for discussion of methodology, validation of results, and interpretation of aerodynamic trends. Feedback sessions and technical discussions contributed to improving the quality of the work, demonstrating basic teamwork and collaborative skills.

PO10 Communication	3	Effectively presented work through: Graphs (C_l , C_d , C_m , C_m vs AOA), Tables and comparisons, PPT and report writing.
P11 Project Management	2	Managed workflow: Geometry → Meshing → Solver → Post-processing, Selection of best configurations, Time-based execution of simulations.
P12 Life-long Learning	2	Self-learned: XFOIL usage, Mesh independence concept, Stability analysis using C_m slope.

MAPPING OF PROGRAM SPECIFIC OUTCOMES

PROGRAM SPECIFIC OUTCOMES:	MAPPING LEVEL:	JUSTIFICATION:
PSO1 Research, Design and Production in Aerospace domain	3	Designed hybrid airfoil geometries using blending technique, performed detailed aerodynamic analysis (C_l, C_d, C_m), Studied stability using C_m vs AOA, compared configurations to identify optimal design (40:60).
PSO2 Competency in Higher Studies and Competitive Exams	3	Build fundamentals in communication systems and simulation.